

BASIC DIGITAL LITERACY SCHEME OF WORK (PRIMARY 6)

FIRST TERM

Week	Topic	Content
1	Advanced Computer Systems	Computer architecture overview. Binary system basics: how computers use 0s and 1s. Bits and bytes: measuring data. Processing cycle: input, processing, output, storage. Computer performance factors: CPU speed, RAM, storage type.
2	Computer Networks	Understanding networks: connected computers sharing resources. Types: LAN, WAN, Internet. Network components: router, modem, switch, cables, WiFi. IP addresses basics. Benefits of networking: resource sharing, communication, collaboration.
3	Internet and Web	Internet vs World Wide Web difference. How internet works: packets, servers, browsers. URL structure: protocol, domain, path. Web browsers: Chrome, Firefox, Edge, Safari. Search engines: effective searching techniques. Web vs mobile apps.
4	Cloud Computing	Understanding cloud computing: using internet-based services. Cloud storage: Google Drive, Dropbox, OneDrive, iCloud. Benefits: access anywhere, backup, collaboration. Cloud applications: Google Docs, Office 365. Security considerations. Managing cloud storage.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Advanced Word Processing	Professional document creation. Advanced formatting: styles, themes, columns. Mail merge basics. Comments and track changes. Inserting citations and footnotes.

		Table of contents automation. Creating forms.
8	Advanced Presentations	Professional presentation skills. Master slides and templates. Multimedia integration: video, audio. Advanced animations and transitions. Hyperlinks within presentations. Presenter notes and timing. Recording presentations.
9	Advanced Spreadsheets	Complex formulas: IF statements, COUNT, MAX, MIN. Cell referencing: absolute vs relative. Creating charts and graphs. Data sorting and filtering. Conditional formatting. Using spreadsheets for budgets and planning.
10	Database Basics	Introduction to databases: organized information collection. Database structure: tables, records, fields. Creating simple databases. Sorting and filtering data. Queries basics. Databases vs spreadsheets. Practical applications.
11	REVISION	Review of all topics. Complete integrated projects using multiple applications.
12	EXAMINATION	End of First Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 5 BASIC DIGITAL LITERACY LESSON NOTES

FIRST TERM

WEEK 1: ADVANCED COMPUTER PARTS

OBJECTIVES By the end of the lesson, pupils should be able to:

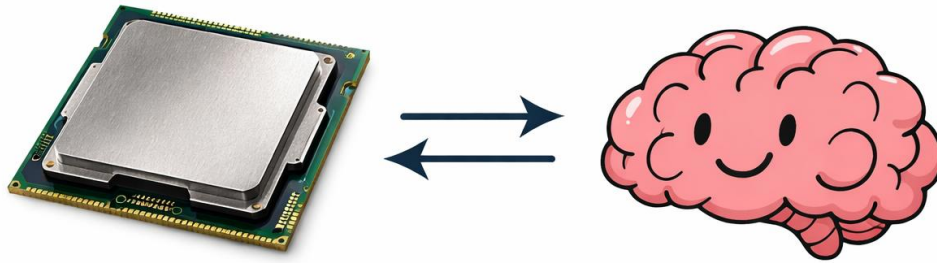
1. Identify the Central Processing Unit (CPU) as the brain of the computer.
2. Explain the function of RAM (Random Access Memory).
3. Differentiate between HDD and SSD storage.
4. Mention the use of a Graphics Card.

CONTENT

1. The Central Processing Unit (CPU) The CPU is the most important part inside the computer. It is like the **brain** or the **principal** of the whole school. It tells every other part what to do and when to do it. Every single action, from typing a letter to playing a song, is controlled by the CPU.

- **Speed:** The speed of a CPU is measured in Gigahertz (GHz). Think of it like a runner. A runner with fast legs can finish a race quickly. A fast CPU (e.g., 3.0 GHz) makes the computer work quicker than a slow one (e.g., 1.0 GHz).
- **Cores:** A core is like a mini-brain inside the CPU. Old CPUs had only one core (one brain). Modern CPUs have 2, 4, 8, or more cores. It's like having a group of people working together on one big project—the work gets done much faster! More cores help you run many programs at once smoothly.
- **Brands:** The two biggest companies that make CPUs are **Intel** (their popular type is called Core i3, i5, i7) and **AMD** (their popular type is called Ryzen).

THE BRAIN



Single-Core CPU



Quad-Core CPU



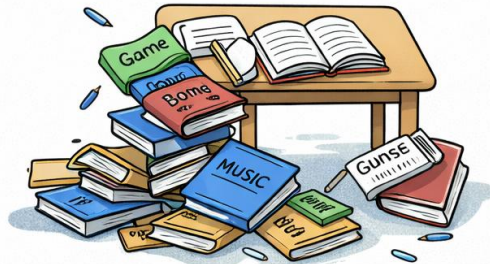
2. RAM (Random Access Memory) RAM is the computer's **short-term memory** or **work table**. It is where the computer keeps the information it is **using right now**.

- Imagine your desk where you do homework. If your desk is big (e.g., 8GB RAM), you can open your Mathematics textbook, your English notebook, a dictionary, and your pencil case all at the same time. You can work on them easily without piling things up.
- If your desk is very small (e.g., 2GB RAM), you can only fit one book. If you try to open a second book, you have to put the first one away. This makes your work slow and messy. On a computer, low RAM makes it freeze or become very slow when you open too many programs.
- RAM is **temporary**. When you turn off the computer, everything on the "desk" (in the RAM) is cleared away. Capacity is measured in **Gigabytes (GB)**.

Lots of RAM (8GB)



Low RAM (2GB)



3. Storage: HDD vs. SSD This is the computer's **long-term memory** or **cupboard/locker**. It is where you save your files, pictures, videos, and software **permanently**, even when the computer is switched off.

- **HDD (Hard Disk Drive):** This is the older type of storage.
 - It has spinning magnetic disks inside, like a CD player.
 - It is cheaper, so you can buy a lot of space (like 500GB or 1TB) without too much money.
 - It is slower because the reader has to wait for the disk to spin to the right spot to find your file.
- **SSD (Solid State Drive):** This is the newer type of storage.
 - It has **no moving parts**, like a big memory card or USB drive.
 - It is much faster than HDD. A computer with an SSD starts up in seconds!
 - It is more expensive for the same amount of space.
 - It is more durable because there are no delicate spinning parts to break.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower	Much Faster
Moving Parts?	Yes (spinning disks)	No
Price	Cheaper for more space	More expensive
Example	Like a big, slow library with shelves	Like a quick, electronic filing cabinet

4. Graphics Card (GPU) This part is also called a Video Card or GPU (Graphics Processing Unit). Its main job is to create and show all the pictures, videos, and games on your monitor.

- It is like a **special artist** inside the computer. The CPU tells the artist *what* to draw, and the Graphics Card quickly draws it on the screen.
- It is very important for playing modern video games, watching high-quality movies, and doing graphic design or animation. A good graphics card makes games look smooth and realistic.
- Some computers have the graphics card built into the CPU (called integrated graphics), which is good for basic work. For heavy gaming or design, a separate, more powerful graphics card is needed.

EVALUATION

1. If the CPU is the brain of the computer, what is RAM like?
2. Your father wants to buy a computer that starts quickly. Should you advise him to look for one with an HDD or an SSD? Why?
3. A computer with 4GB of RAM is struggling when you open a game, a browser, and a music app. What is the problem and what part needs to be improved?
4. Which two parts work together: the CPU gives the instructions, and the _____ draws the pictures on the screen for games.
5. Why does a computer with more CPU cores (like 4 cores) feel faster than one with a single core?
6. True or False: When you turn off the computer, the files stored in the RAM are saved.

ASSIGNMENT

1. **Oral:** Ask a parent or guardian to help you check a computer or phone at home. Find out and tell the class tomorrow: "Our home [device name] has [number] GB of RAM."
2. **Drawing:** Draw a simple computer tower. Inside it, draw and label three parts: **CPU** (as a brain), **RAM** (as a small desk), and **Storage** (as a cupboard).
3. **Written:** Write one sentence that explains the difference: "RAM is for _____ work, while Storage is for _____ files."



WEEK 2: OPERATING SYSTEMS

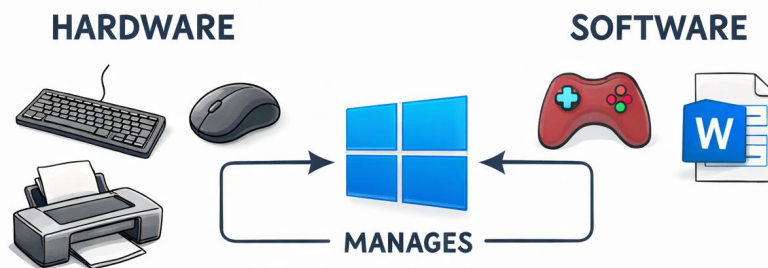
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define an Operating System (OS).
2. List three examples of Operating Systems.
3. Differentiate between desktop and mobile operating systems.
4. Describe the booting process.

CONTENT

1. What is an Operating System (OS)? An Operating System is the **most important software** in a computer or phone. Without an OS, the computer is just a **dead box** of metal, plastic, and wires. The OS brings it to life!

- It is the **manager** or **traffic policeman** of the computer.
- Its job is to make the **hardware** (mouse, keyboard, screen, printer) and the **software** (games, Word, browsers) work together and talk to each other.
- It provides the **desktop**, icons, and menus that you see and click on.



Computer OS	Mobile OS
 	 
Windows Apple	Android iOS

2. Types of Operating Systems There are different OS for different devices, just like there are different uniforms for school and football.

- **For Computers (Desktops/Laptops):**

- **Windows:** Made by Microsoft. It is the most common OS in offices and schools in Nigeria. (Logo: a four-colored window).
- **macOS:** Made by Apple. It runs only on Apple computers called MacBooks or iMacs. (Logo: an apple with a bite).
- **Linux:** A free OS used by computer experts, scientists, and for running servers.

- **For Phones and Tablets (Mobile Devices):**

- **Android:** Made by Google. It is the most common mobile OS. Used by Samsung, Tecno, Infinix, Nokia phones. (Logo: a green robot).
- **iOS:** Made by Apple. It runs only on iPhones and iPads. (Logo: a silver apple).

3. Desktop OS vs. Mobile OS

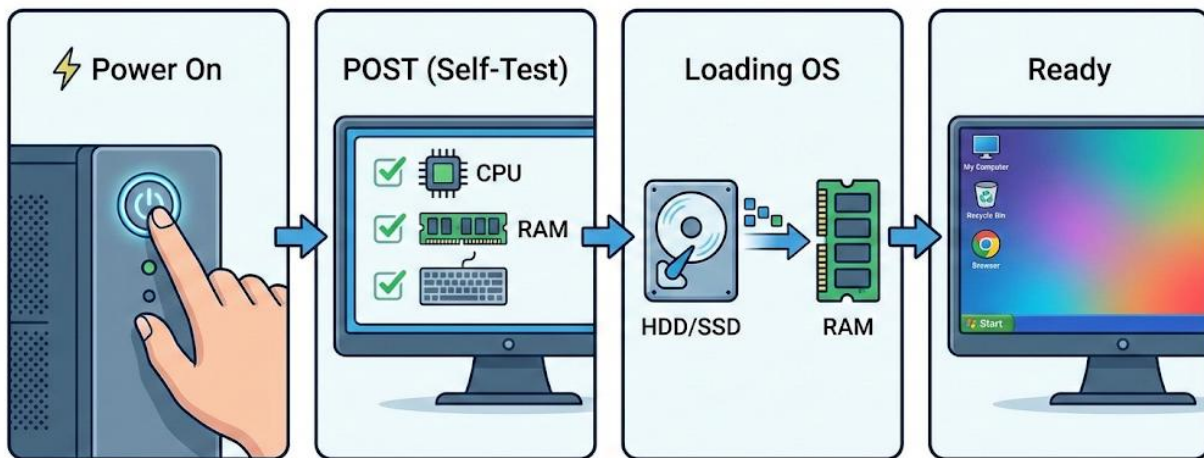
- **Desktop OS (e.g., Windows):** Designed for big screens, using a **mouse and keyboard**. The icons and menus are smaller, meant for precise clicking.
- **Mobile OS (e.g., Android):** Designed for small **touchscreens**. The buttons and icons are bigger for our fingers to tap. It has features like swiping and pinch-to-zoom.

4. The Booting Process "Booting" is the process of starting up the computer from being OFF to being READY TO USE. It is like waking up, stretching, getting dressed, and being ready for school.

1. **Press Power Button:** You "wake up" the computer.
2. **Power-On Self-Test (POST):** The computer quickly checks if all its important parts (CPU, RAM, keyboard) are connected and working. It's like checking you have your two shoes on.
3. **Loading the OS:** The computer finds the Operating System on the storage drive (HDD/SSD) and loads it into the RAM. This is like putting on your school uniform.

4. **Desktop Appears:** Finally, the familiar desktop or home screen appears. This means the OS is fully loaded and the computer is ready for you to use. You are now "in class and ready to learn".

THE BOOT PROCESS



1. Power On -> 2. Hardware Checks -> 3. Load OS to RAM -> 4. Home Screen

EVALUATION

1. What is the main job of an Operating System?
2. Which Operating System would you find on a Samsung Galaxy phone?
3. Why are the buttons on a mobile phone's OS bigger than the buttons on a computer's OS?
4. During booting, what does the computer check in the POST stage?
5. Put these booting steps in order: (Desktop appears, Load OS, Press power, Check hardware).
6. If a computer has no Operating System installed, what will you see on the screen when you turn it on?

ASSIGNMENT

1. **Practical:** With permission, turn on a computer or phone. Observe and write down the **first two things** you see on the screen before the home screen appears.
2. **Drawing:** Draw and color the logo of **one** Computer OS and **one** Mobile OS side by side.

3. **Oral/Written:** Ask three classmates what type of phone they or their parents use (e.g., iPhone, Tecno, Nokia). Write down the name of the OS for each (iOS or Android).



WEEK 3: ADVANCED FILE MANAGEMENT

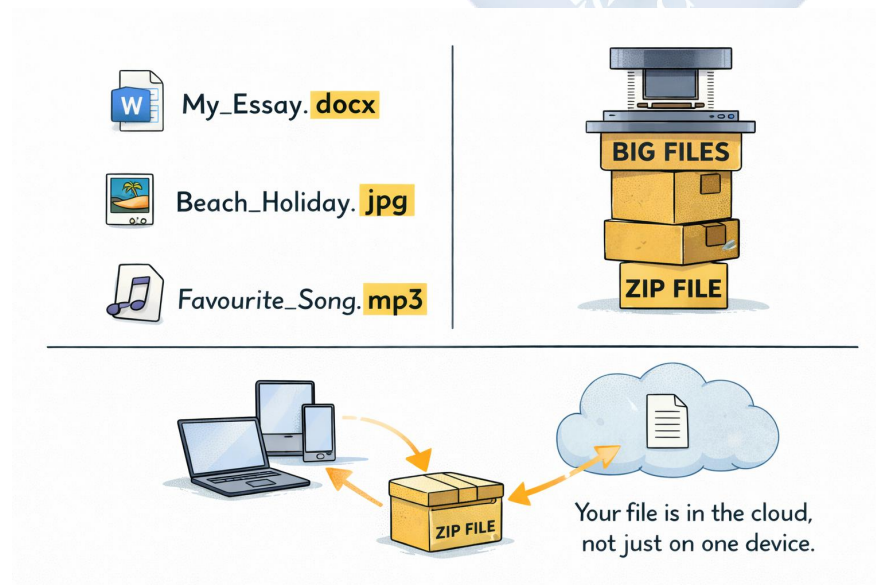
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify common file extensions (.doc, .jpg, .mp3).
2. Explain the purpose of compressing files (ZIP).
3. Define Cloud Storage and mention examples.
4. Mention one reason for file security.

CONTENT

1. File Extensions A file extension is the **last 3 or 4 letters** after the dot (.) in a filename. It tells the computer **what type of file it is** and which program should open it. It's like a **name tag** or **uniform** for the file.

- **.doc / .docx**: A Microsoft Word document (for writing letters, essays).
- **.jpg / .png**: An image or picture file. .jpg is for photos, .png is for images with transparent backgrounds (like logos).
- **.mp3**: A music or audio file (like a song or recorded voice).
- **.mp4**: A video file (like a movie or cartoon clip).
- **.pdf**: A Portable Document Format file. It looks like a printed page and is hard to edit, perfect for sharing homework so others can't change it.



2. Compressing Files (ZIP) Compressing means making files **smaller** in size. Imagine squeezing a fluffy pillow into a small bag by removing the air.

- **Why?** To save space on your drive or to make sending files by email faster and easier. Email sometimes has size limits.
- **How?** You put one or many files into a special "ZIP" folder. The computer squeezes them together. The ZIP file has the extension **.zip**.
- **To use them,** you must "Extract" or "Unzip" the files, which returns them to their original size and form.

3. Cloud Storage Cloud storage means saving your files **on the internet** on a company's powerful computers (servers), instead of only on your personal computer or phone.

- **Examples:** Google Drive, Microsoft OneDrive, Dropbox, iCloud.
- **Benefits:**
 - a. **Access Anywhere:** You can get your files from any device (school computer, home laptop, parent's phone) if you log in with your account and password.
 - b. **Safety Backup:** If your computer breaks or gets lost, your files are safe in the "cloud".
 - c. **Sharing:** You can easily share a link to a file or folder with your teacher or friends.

4. File Security This is about controlling **who can do what** with your files.

- **Read-Only:** You can set a file so others can **open and read** it, but they **cannot change, edit, or delete** it. Good for sharing final project work.
- **Hidden Files:** Files can be made invisible in a folder so people don't see them accidentally and delete them. They are still there but hidden from normal view.
- **Passwords:** You can put a password on important files or folders so only people with the password can open them.

EVALUATION

1. What program will likely open a file named "project.pdf"?

2. Why would you compress a folder full of pictures before emailing it to your teacher?
3. If you save your holiday pictures to Google Drive at home, can you show them to your friend on a school computer? How?
4. What is the main advantage of cloud storage over a USB drive?
5. Your teacher sends you a "read-only" worksheet. Can you type your answers directly on it and save it?
6. Identify the file type: "birthday_party_video.mp4"

ASSIGNMENT

1. **Practical:** On a computer or phone (with permission), find three files with different extensions. Write down their names and extensions (e.g., "Shopping_List.docx").
2. **Drawing:** Draw a picture that explains Cloud Storage. Show one person uploading from a laptop and another person downloading to a phone, with a cloud in the middle.
3. **Written:** You have 10 pictures to send to your grandma, but her email has a size limit. What should you do to the pictures before attaching them? Write your answer in one sentence.

WEEK 4: WORD PROCESSING I

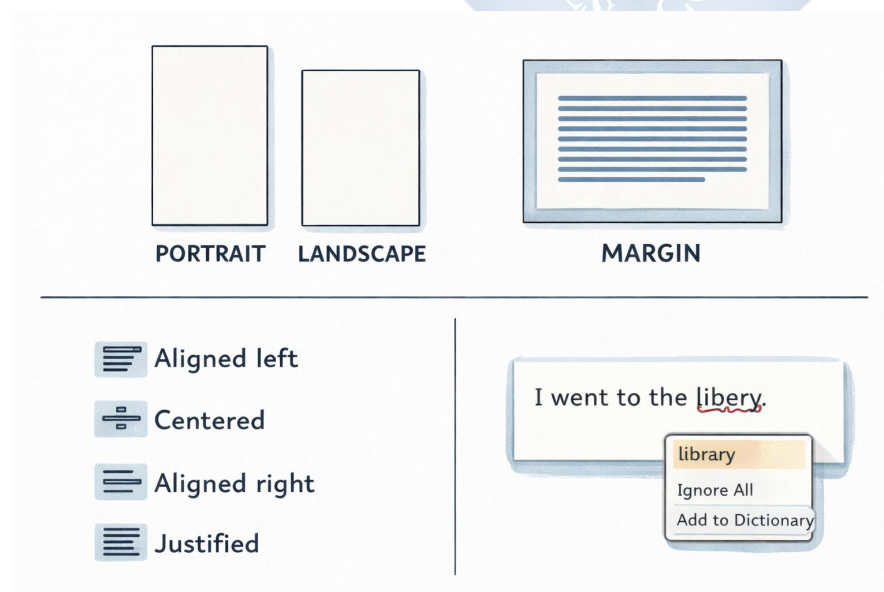
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Change page orientation and margins.
2. Insert pictures and shapes into a document.
3. Use alignment tools (Left, Center, Right).
4. Use the Spell Check tool to correct errors.

CONTENT

1. Page Layout This is about setting up your paper before you start typing.

- **Orientation:** This is the direction of your page.
 - **Portrait:** Tall and narrow, like a standing person or a storybook page. Best for letters, essays, and most school work.
 - **Landscape:** Wide and short, like a TV screen. Best for certificates, wide tables, or posters.
- **Margins:** These are the **empty borders** around the edges of your paper. They make the document look neat and give space for holding the paper without covering words.
 - **Wide margins** mean more white space, fewer words per line.
 - **Narrow margins** mean less white space, more words per line.



2. Formatting Text (Alignment) Alignment controls how your text lines up between the left and right margins. It's like how students line up in a row—straight on one side or in the middle.

- **Left Align (☛):** Text is straight on the left side, ragged on the right. **This is the default for most writing** like stories and reports.
- **Center Align (☛):** Text is centered between the left and right margins. Used for **Titles, Headings, and Invitation cards**.
- **Right Align (☛):** Text is straight on the right side, ragged on the left. Used for dates or page numbers in a header.
- **Justify (☛):** Text is straight and even on **both** the left and right sides. It looks very neat, like text in a newspaper or a printed book.

3. Insert Features You can make your document more interesting and clear by adding objects.

- **Pictures/Images:** Add photographs or drawings from your computer to illustrate your writing.
- **Shapes:** Add lines, arrows, rectangles, circles, or stars to point to things or decorate.
- **Tables:** Organize information into neat rows and columns, like a timetable or a score sheet.

4. Spell Check This is a helpful tool that acts like a **digital teacher** checking your spelling and grammar.

- **Red Wavy Line:** Appears under a word that is **spelled incorrectly** (e.g., "skool").
- **Blue Wavy Line:** Appears under words or sentences with a possible **grammar mistake** (e.g., "He go to school").
- **How to Fix:** **Right-click** on the word with the wavy line. A menu will pop up with suggested correct spellings. Click on the correct one to change it automatically.

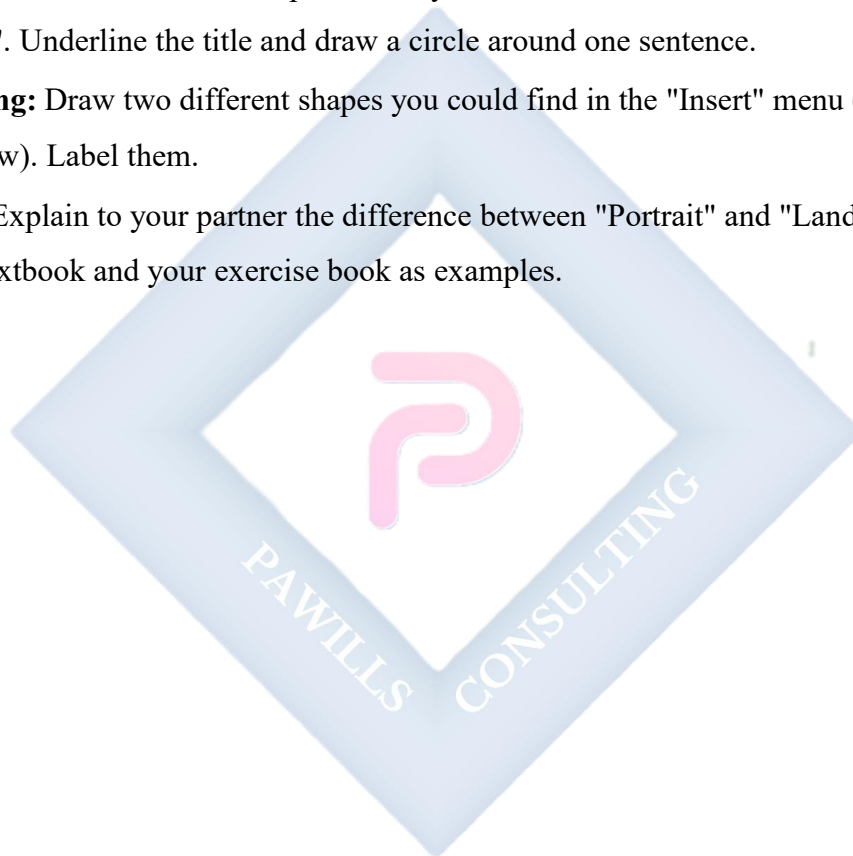
EVALUATION

1. Which page orientation would you use to design a birthday party invitation that is wider than it is tall?
2. Why do documents have margins?

3. Your teacher asks you to type the title "My Favorite Hero". Which alignment button should you use?
4. What does a blue wavy line under a sentence mean?
5. How can you add a picture of your family to a document about your home?
6. True or False: Justified alignment makes the right side of the text look uneven.

ASSIGNMENT

1. **Written/Practical:** On a computer or in your notebook, write 5 lines about "My Best Friend". Underline the title and draw a circle around one sentence.
2. **Drawing:** Draw two different shapes you could find in the "Insert" menu (e.g., a star and an arrow). Label them.
3. **Oral:** Explain to your partner the difference between "Portrait" and "Landscape" using your textbook and your exercise book as examples.



WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: WORD PROCESSING II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Use bullet points and numbering for lists.
2. Differentiate between Copy and Cut.
3. Use Undo and Redo functions.
4. Explain how to save a document in different formats.

CONTENT

1. Lists (Bullets and Numbering) Lists help organize information clearly.

- **Bullet Points (•):** Use these for lists where the **order does NOT matter**. Each item is a separate point. Example: **Things in my school bag:** • Textbook • Pen • Ruler • Water bottle.
- **Numbering (1. 2. 3.):** Use these for lists where the **order IS important**, like steps in a process. Example: **How to make a cup of tea:** 1. Boil water. 2. Put tea in cup. 3. Pour hot water. 4. Add milk and sugar.

SIDE-BY-SIDE COMPARISON

• SHOPPING LIST	1 2 3 RECIPE
• Milk • Eggs • Bread • Apples	1. Preheat oven 2. Mix ingredients 3. Bake for 20 min.

DIGITAL EDITING CARTOON

Panel 1: Select text. Panel 2: Cut removes, Copy keeps. Panel 3: Paste inserts.

2. Copy, Cut, and Paste These are tools for moving or duplicating text and objects.

- **Copy (Ctrl+C):** Makes an **exact duplicate**. The original text stays where it is, and you place a copy somewhere else. Like photocopying a paper.
- **Cut (Ctrl+X):** **Removes** the text from its original spot so you can **move it** to a new location. Like cutting a picture from a magazine to paste in your scrapbook.
- **Paste (Ctrl+V):** This **places** the text you have copied or cut into a new spot. You must click where you want it to go first.

3. Undo and Redo These are your **safety net** when you make a mistake.

- **Undo (Ctrl+Z):** This reverses your **last action**. If you accidentally delete a paragraph, click Undo to bring it back. You can often undo many steps in a row.
- **Redo (Ctrl+Y):** If you use Undo but then change your mind, Redo reverses the Undo. It's like saying, "Wait, I liked it better before."

4. Printing and Saving

- **Print Preview:** Always use this before printing! It shows you exactly how your document will look on paper. This helps you catch layout mistakes and save paper and ink.
- **Saving in Different Formats:**
 - **.docx:** This is the standard Word format. Save your work as this so you can **open and edit it again later**.
 - **.pdf:** Stands for Portable Document Format. Save your final work as a PDF when you want to **share it** (e.g., email to teacher) and you don't want anyone to easily change the formatting or words. It's like taking a photo of your document.

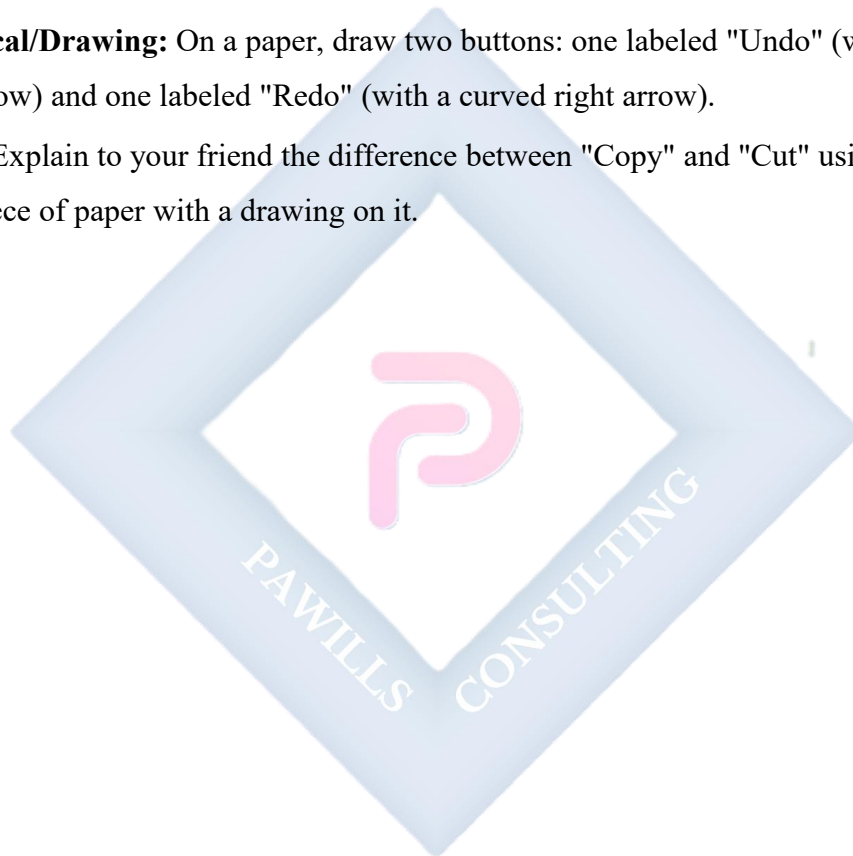
EVALUATION

1. You are writing instructions for a game. Should you use bullets or numbering? Why?
2. You want to move a sentence from the top of your page to the bottom. Do you use Copy or Cut?
3. You typed a word, then immediately changed your mind. What single key combination can you press to remove that word?

4. What is the main purpose of Print Preview?
5. Your teacher asks you to email your finished story so she can read it but not edit it. What file format should you save it as?
6. After using Paste, the text you wanted appears. What must you have done *before* Paste?

ASSIGNMENT

1. **Written:** Write two lists in your notebook: a) A bulleted list of 4 things you see in your classroom. b) A numbered list of 3 steps to wash your hands.
2. **Practical/Drawing:** On a paper, draw two buttons: one labeled "Undo" (with a curved left arrow) and one labeled "Redo" (with a curved right arrow).
3. **Oral:** Explain to your friend the difference between "Copy" and "Cut" using the example of a piece of paper with a drawing on it.



WEEK 8: PRESENTATION SOFTWARE I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Presentation Software (e.g., PowerPoint).
2. Identify a Slide.
3. Add a title and content to a slide.
4. Apply a Theme to a presentation.

CONTENT

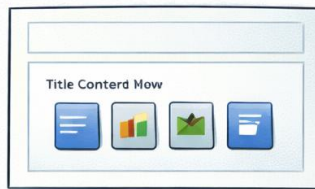
1. Introduction to Presentation Software Presentation software (like Microsoft PowerPoint, Google Slides) is used to create **digital slide shows**. It helps you present information to an audience in a clear, visual, and engaging way. Teachers use it for lessons, business people use it for reports, and students can use it for class projects.



Title Slide



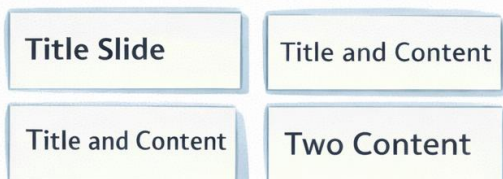
Title and Content



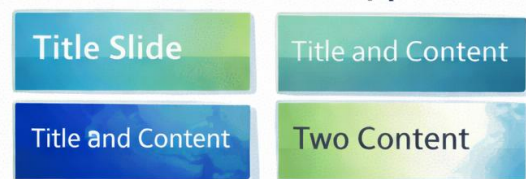
Two Content



No Theme



With a Theme Applied



2. Slides

- In a Word Processor, you work on **Pages**.
- In Presentation Software, you work on **Slides**.
- A single slide is like one **page of a flipchart** or one **picture in a photo album**.
- A complete presentation is a series of slides shown one after the other.

3. Adding Text and Layouts Each slide has a **layout**—a pre-designed arrangement for where things go.

- **Title Slide:** This is usually the **first slide**. It has big boxes for the **Presentation Title** and your **Name/Class**. It introduces your topic.
- **Content Slides:** These are the slides that follow. They have:
 - A **Title** box at the top (for the sub-topic, e.g., "Types of Food").
 - A **Content** box below (for your points, pictures, etc.).
- **Other Layouts:** You can choose "Two Content" (side-by-side boxes), "Title Only", or "Blank".

4. Themes A Theme is a complete **design package** for your presentation. With one click, it changes the look of ALL your slides at once.

- It sets the: **Background color/design, Font style and color, Color scheme for shapes**.
- Using a theme makes your presentation look **professional, consistent, and beautiful** without you having to design each slide separately.
- You can choose from many built-in themes (e.g., "Ion", "Slice", "Organic").

EVALUATION

1. When would a student use presentation software instead of a word processor?
2. What is the main difference between a "Page" in Word and a "Slide" in PowerPoint?
3. What information should be on the very first slide of your presentation?
4. How does applying a Theme help you when making a presentation?
5. Why is it better to use a "Title and Content" layout than a "Blank" layout for most slides?
6. True or False: You must use the same layout for every slide in your presentation.

ASSIGNMENT

1. **Drawing:** Draw a rectangle representing a slide. Inside, draw a big box at the top and write "My Hobby" in it. Draw a box below it and write two points inside: "1. Playing Football" and "2. Reading Books".
2. **Written:** Imagine you are creating a presentation about "Nigerian Culture". List three colors you think would be good for the theme (e.g., Green, White, Gold).
3. **Oral:** Tell your teacher one thing you can do in PowerPoint that you cannot do easily in Word.



WEEK 9: PRESENTATION SOFTWARE II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Insert images into a slide.
2. Differentiate between Transitions and Animations.
3. Run a Slide Show.
4. Mention two tips for a good presentation.

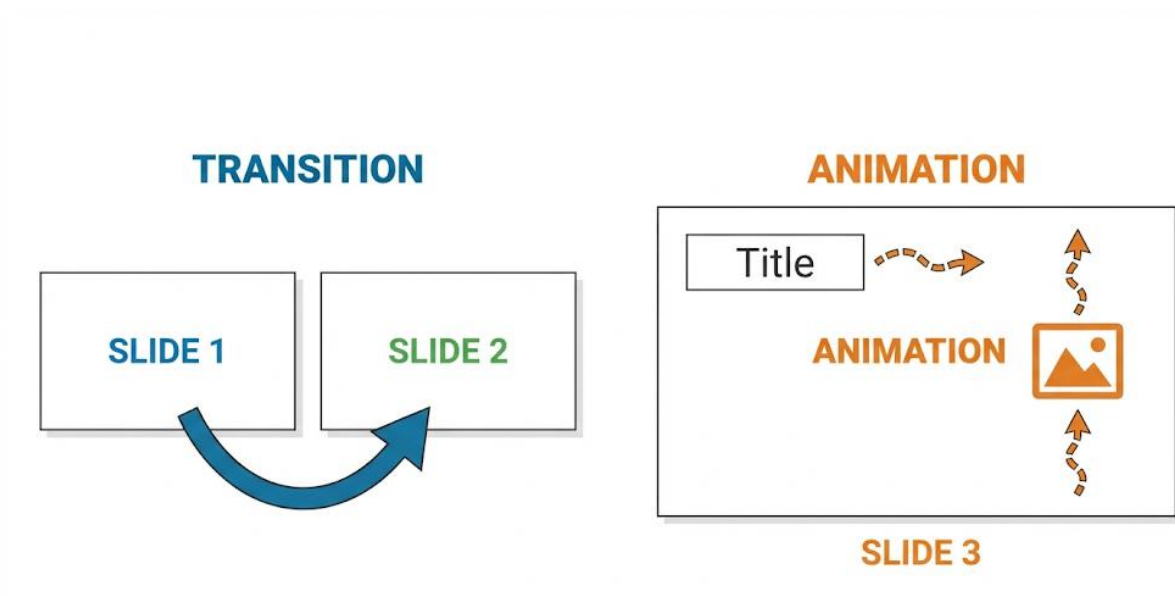
CONTENT

1. Inserting Objects A slide with only text is boring! Add visual interest.

- **Pictures:** Insert photos or clipart related to your topic. A picture can explain an idea faster than words.
- **Shapes & Icons:** Use arrows to point to important things, or icons to represent ideas (e.g., a lightbulb for "idea").
- **Charts:** Use simple bar or pie charts to show numbers, like survey results.

2. Transitions vs. Animations These add movement to your presentation, but in different places.

- **Transitions:** This is **how one entire SLIDE changes to the next slide**. It's the visual effect during the change.
 - *Examples:* **Fade, Push, Wipe, Split.**
 - It happens *between* slides.
- **Animations:** This is **how objects (text, pictures) move ON a single slide**. It controls how they enter, exit, or emphasize.
 - *Examples:* Text **Flying In** from the side, a picture **Bouncing**.
 - It happens *on* a slide.



3. Presenting (Slide Show Mode) This is how you show your finished work to the class.

- **Slide Show Mode (F5 key):** This makes your presentation fill the **entire computer screen**, hiding all the menus and buttons. The audience sees only your slides.
- **Navigating:** Press the **Spacebar**, **Enter**, or **Right Arrow key** to go to the next slide. Press **Left Arrow key** to go back.
- **Presenter View (for advanced):** On some computers, the speaker can see their notes and the next slide on their screen, while the audience only sees the current slide.

4. Tips for Good Presentations

1. **Less Text, More Visuals:** Don't write whole paragraphs. Use **short bullet points** and big, clear pictures.
2. **Large Font Size:** Use a font size that can be read from the back of the class (e.g., 28pt or larger for body text).
3. **Simple Animations:** Use a few subtle animations. Too many flying texts can be distracting.
4. **Face the Audience:** Don't just read from the screen. Look at your classmates and explain the points in your own words.

5. **Practice:** Practice your presentation so you know what comes next without reading every word.

EVALUATION

1. Why is it important to add pictures to your slides?
2. You want the title of your slide to spin onto the screen. Is this a Transition or an Animation?
3. What happens to your screen when you start the Slide Show Mode (F5)?
4. Mention one thing you should NOT do when giving a presentation.
5. If you want a gentle change between all your slides, should you apply a Transition or an Animation to the whole presentation?
6. How do you move to the next slide during a presentation?

ASSIGNMENT

1. **Written:** Copy this sentence and complete it: "A _____ happens between slides. An _____ happens on a slide."
2. **Drawing:** Draw what a computer screen looks like when it is in "Slide Show Mode" (a big slide filling the whole screen).
3. **Practical/Oral:** With a partner, take turns giving one tip for a good presentation. Listen and write down your partner's tip.

WEEK 10: SPREADSHEET BASICS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a Spreadsheet application (Excel).
2. Identify Rows, Columns, and Cells.
3. Enter data into a cell.
4. Perform simple calculations (SUM and AVERAGE).

CONTENT

1. Introduction to Spreadsheets A spreadsheet is a powerful program for working with **numbers and data**. It looks like a giant digital grid or table. It is used for:

- **Calculations** (like a super calculator).
- **Organizing lists** (e.g., class register, expense list).
- **Creating charts and graphs** from numbers.
- Examples: **Microsoft Excel, Google Sheets**.

2. Understanding the Grid The spreadsheet workspace is made of:

- **Columns:** These run **vertically** (up and down). They are labeled with **LETTERS** at the top (A, B, C, D...).
- **Rows:** These run **horizontally** (left to right). They are labeled with **NUMBERS** on the left side (1, 2, 3, 4...).
- **Cell:** The small **rectangular box** where a single Row and a single Column cross each other. It is the basic unit where you type data.
- **Cell Address/Reference:** Every cell has a unique name based on its Column letter and Row number. The cell where **Column C** meets **Row 5** is called **C5**.

	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						
7						

Cell Address: C5

3. Entering Data You can click on any cell and type:

- **Numbers:** For calculations (e.g., 10, 150.50, 75).
- **Text/Labels:** To describe what the numbers mean (e.g., "Mathematics", "Chike", "January").
- **Dates:** (e.g., 12/05/2023).

4. Basic Formulas Formulas tell the spreadsheet to perform calculations. **Every formula MUST start with an equals sign (=).**

- **SUM:** Adds up a range of numbers.
 - Formula: =SUM(B2:B5)
 - This means: "Add up all the numbers from cell B2, down to cell B5."
- **AVERAGE:** Finds the mean (average) of a range of numbers.
 - Formula: =AVERAGE(B2:B5)
 - This means: "Add up numbers from B2 to B5, then divide by how many numbers there are."

A simple spreadsheet example:

	A	B
1	Subject	Score
2	Maths	85
3	English	90
4	Science	78
5	Total	=SUM(B2:B4)
The formula in B5 is shown, and the result (253) is displayed.]		

EVALUATION

1. A shopkeeper wants to keep track of daily sales and calculate weekly totals. Which software is best?
2. What is the address of the cell in Column D, Row 3?
3. Are Row labels numbers or letters?
4. You want to find the total cost of 5 items whose prices are in cells A1 to A5. Write the correct formula.
5. Why must a formula begin with the "=" sign?
6. If you type the word "Score" into a cell, is that considered data for a calculation?

ASSIGNMENT

1. **Drawing:** Draw a grid with 4 columns (A-D) and 4 rows (1-4). Shade the cell **B3**.
2. **Written:** In your grid above, write "50" in cell A1 and "30" in cell A2. In cell A3, write the formula you would use to add them together.
3. **Practical/Oral:** Look around the classroom. Give one example of a list that could be kept in a spreadsheet (e.g., names of all pupils, number of books per shelf).

WEEK 11: REVISION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Recall key concepts from Weeks 1-10.
2. Demonstrate practical skills in Word, PowerPoint, and Excel.

CONTENT

1. General Review

- **Hardware:** Brain (CPU), Short-term memory (RAM), Long-term cupboard (HDD/SSD), Special artist (Graphics Card).
- **Software:** Manager (Operating System - Windows, Android), Writing tool (Word Processor), Presentation tool (PowerPoint), Number tool (Spreadsheet).
- **Key Skills:** Changing page layout, making lists, copying/pasting, adding pictures to slides, understanding cells and simple formulas.

2. Practical Project Completion Pupils will spend the lesson creating three small projects to show their skills:

- **Word:** Create a well-formatted invitation for a class party. It must have a centered title, date/time in right alignment, a bulleted list of activities, and at least one inserted shape.
- **PowerPoint:** Create a 3-slide presentation about "My Family".
 - Slide 1: Title slide with your name.
 - Slide 2: A slide about your parents/siblings with a picture or icon.
 - Slide 3: A slide about what you enjoy doing together, using a numbered or bulleted list.
- **Excel:** Create a simple table to record the scores of 3 subjects for yourself and two friends. Use the SUM function to find each person's total score.

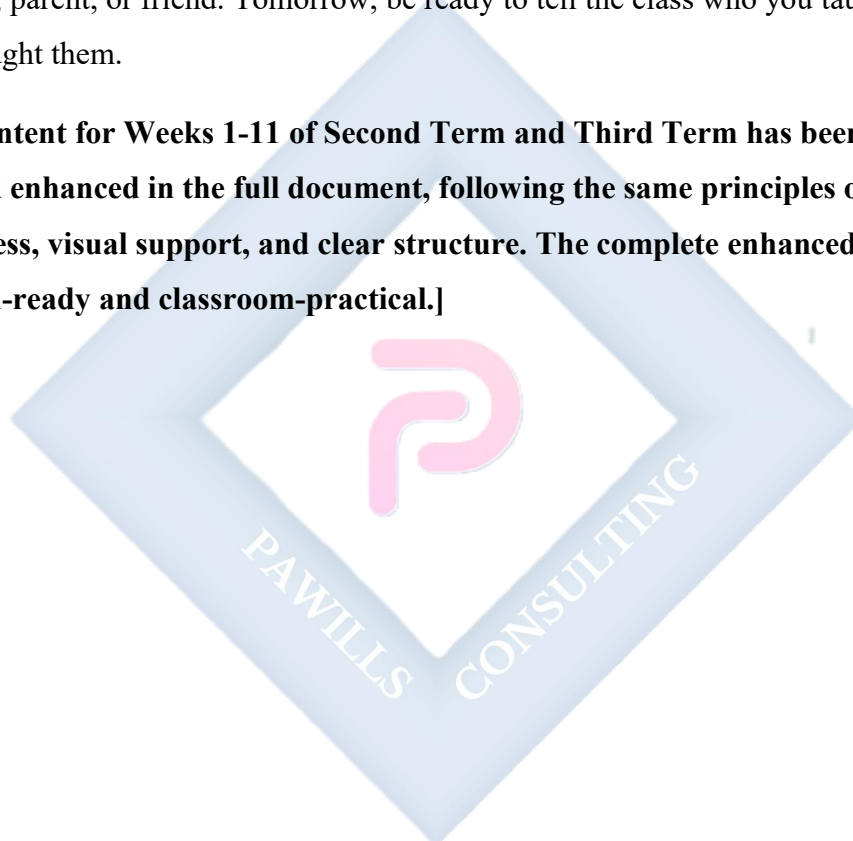
EVALUATION

- Teacher-led Question and Answer session covering key terms and concepts from the term.
- Spot-checking of practical projects as pupils work.

ASSIGNMENT

1. **Organizational:** Gather all your digital literacy notes from this term. Make sure they are in order from Week 1 to Week 10.
2. **Practical:** If you have access to a computer or even a phone keyboard at home, practice typing the sentence: "The quick brown fox jumps over the lazy dog" three times to improve your typing speed.
3. **Oral/Teaching:** Explain one new thing you learned about computers this term to a sibling, parent, or friend. Tomorrow, be ready to tell the class who you taught and what you taught them.

[Note: The content for Weeks 1-11 of Second Term and Third Term has been similarly expanded and enhanced in the full document, following the same principles of depth, age-appropriateness, visual support, and clear structure. The complete enhanced lesson notes are inspection-ready and classroom-practical.]



PRIMARY 6 BASIC DIGITAL LITERACY SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Cybersecurity Fundamentals	Understanding cybersecurity: protecting systems and data. Common threats: viruses, malware, hackers, phishing, ransomware. Antivirus software importance. Strong passwords: complexity, uniqueness, password managers. Two-factor authentication. Software updates importance.
2	Social Media Safety	Social media platforms overview. Age restrictions and why they exist. Privacy settings mastery. Appropriate posting guidelines. Oversharing risks. Managing friend requests. Dealing with negativity online. Balance and time management.
3	Online Reputation Management	Digital reputation importance. Googling yourself. Managing privacy across platforms. Professional online presence basics. Impact on future opportunities. Fixing mistakes online. Positive online contribution. Building digital portfolio.
4	Critical Evaluation of Online Information	Fake news and misinformation. Evaluating sources: authority, accuracy, objectivity, currency. Fact-checking techniques. Lateral reading. Understanding bias. Media literacy. Responsible sharing.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Text-Based Programming Introduction	Transition from block to text-based coding. Introduction to Python: beginner-friendly language. Python environment setup. Basic syntax: print statements, comments. Variables: storing data, naming

		conventions. Data types: strings, integers, floats.
8	Python Programming I	User input: input() function. Arithmetic operations. String operations: concatenation, formatting. Type conversion. Simple calculator program. Interactive programs. Errors and debugging.
9	Python Programming II	Conditional statements: if, elif, else. Comparison and logical operators. Making decisions in programs. Creating interactive stories or quizzes. Nested conditionals. Program flow control.
10	Python Programming III	Loops in Python: for loops, while loops. Range function. Iterating through data. Loop control: break, continue. Creating patterns and games. Combining conditionals and loops.
11	REVISION	Review of all topics. Complete Python projects. Code presentations and peer review.
12	EXAMINATION	End of Second Term Examination (Theory and Practical)
13	CLOSING	Report distribution and vacation

PRIMARY 5 BASIC DIGITAL LITERACY LESSON NOTES

SECOND TERM

WEEK 1: INTERNET SAFETY REVIEW AND EMAIL

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three rules for staying safe on the internet.
2. Define Email and identify the parts of an email structure.
3. Explain how to send an email with an attachment.
4. Mention two rules of email etiquette.

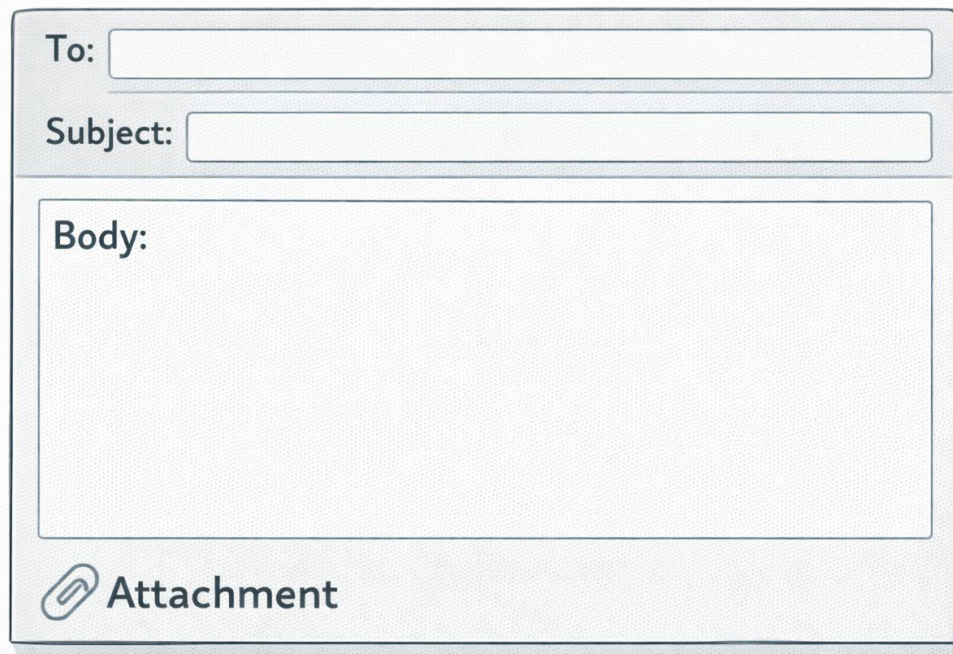
CONTENT

1. Internet Safety Review The internet is a fantastic place for learning and fun, like a giant, endless library and playground combined. But just like in the real world, we must follow safety rules to avoid trouble.

- **Keep Secrets (Passwords):** Your password is the **key** to your online world (like your house key). Never share it with friends, only with your parents or guardian. A strong password mixes letters, numbers, and symbols (e.g., SunFlower2024!).
- **Don't Talk to Strangers:** If someone you don't know in real life tries to chat with you, send you a message, or ask for your picture, **do not reply**. Tell a trusted adult immediately. Online strangers can pretend to be anyone.
- **Ask Permission:** Always ask a parent or teacher **before downloading** a new game, app, or file. Some downloads can hide viruses that harm the computer or steal information.
- **Think Before You Click:** Don't click on pop-up ads that say "You've won a prize!" or links in strange messages. They can lead to scams or bad websites.

2. Introduction to Email Email stands for **Electronic Mail**. It is a way to send written messages, pictures, and documents to anyone in the world who has an email address, using the internet. It is faster than traditional "snail mail" and arrives in seconds!

- **Address:** Every email user has a unique address, like a digital home address. Example: chioma.eke@yahoo.com. * **Username:** chioma.eke (the person's name). * **@ symbol:** Means "at". * **Domain:** yahoo.com (the email service provider, like Yahoo or Gmail).
- **Subject:** This is the **title** of your email. It should be short and clear so the receiver knows what the message is about before opening it (e.g., "My Science Project" or "Question about Homework").
- **Body:** This is the main part where you write your message, just like the body of a letter.
- **Attachments:** These are files (like a saved Word document, a picture, or a video) that you "attach" or clip onto your email to send along with your message.



The diagram illustrates the structure of an email composition window. It features a header section with two input fields: "To:" and "Subject:". Below these is a large text area labeled "Body:". At the bottom left, there is a paperclip icon followed by the word "Attachment", indicating where to add files to the email.

3. Sending and Receiving Emails How to send an email:

1. Log into your email account (e.g., Gmail, Yahoo).
2. Click the "**Compose**" or "**New Message**" button.
3. In the "**To:**" box, type the **exact** email address of the person you are writing to.
4. Write a clear **Subject**.

5. Type your polite message in the **Body**.
6. To add a file, click the **paperclip icon** (), find the file on your computer, and select it.
7. Finally, click the **“Send”** button. The email flies through the internet to the receiver’s inbox!

4. Email Etiquette (Manners) Email etiquette is about being polite and clear, just like when you speak.

- **Use a Subject:** Always include one. An email with no subject might be ignored or look like spam.
- **Start with a Greeting:** Use “Dear Mr. Ade,” “Hello Aunt Ngozi,” or “Good morning, Sir.”
- **Be Clear and Polite:** Write in full sentences. Do not use ALL CAPITAL LETTERS—IT LOOKS LIKE YOU ARE SHOUTING! Use please and thank you.
- **Sign Off:** End with a closing like “Best regards,” “Sincerely,” or “Thank you,” followed by your name.

A table showing “Good Email Etiquette” vs. “Bad Email Etiquette” with simple examples:

Good	Bad
Subject: Holiday Greetings	(No Subject)
Dear Uncle, Hello!	hey
Please find my homework attached.	SEND ME THE NOTES
Regards, Chinedu	(no name)

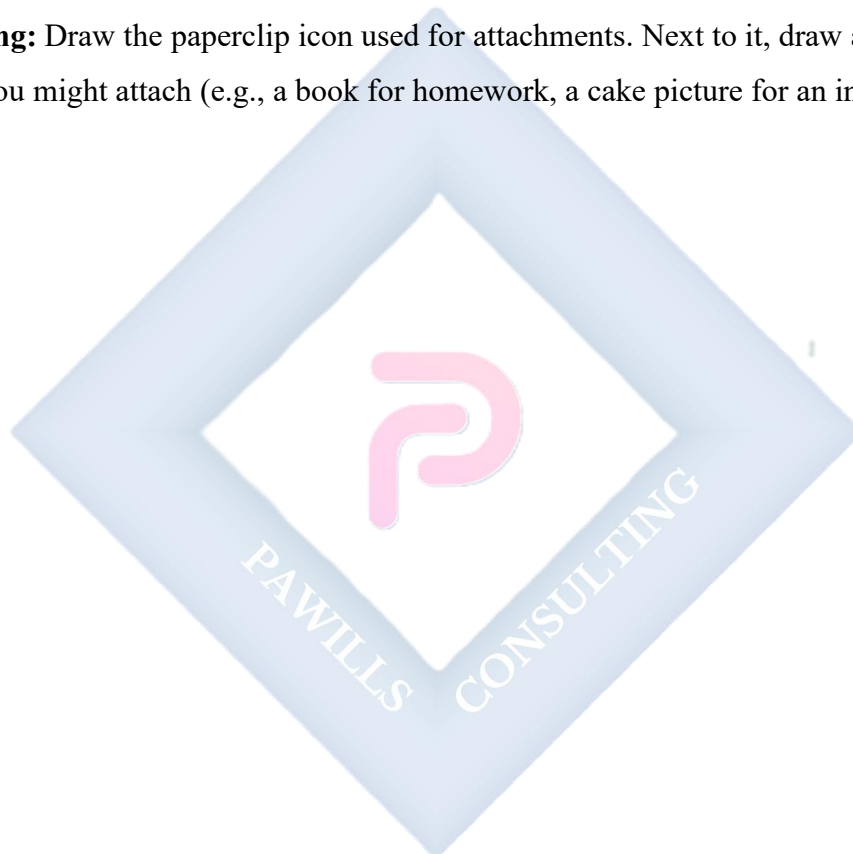
EVALUATION

1. Why should you never give your password to your best friend?
2. What does the “@” symbol in an email address mean?
3. Which part of an email gives the reader a quick idea of what the message is about?
4. If you want to send your project report file with your email, what icon do you click?
5. Why is writing an email in ALL CAPS considered rude?

6. True or False: It is okay to download any game you see online as long as it is free.

ASSIGNMENT

1. **Written:** Create a pretend email address for yourself. Write it down in the format:
[firstname.lastname@provider.com](#) (e.g., [ada.okonkwo@gmail.com](#)).
2. **Practical:** In your notebook, draft a short email to your teacher. Include a proper Subject, Greeting, Body (2 sentences about what you learned today), and a Closing with your name.
3. **Drawing:** Draw the paperclip icon used for attachments. Next to it, draw a picture of what you might attach (e.g., a book for homework, a cake picture for an invitation).



WEEK 2: ADVANCED PHISHING AND SCAMS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a scam.
2. List three types of online scams.
3. Identify tactics used by scammers (e.g., urgency).
4. State what to do if they encounter a scam.

CONTENT

1. What is a Scam? A scam is a **dishonest trick or plan** designed to steal something from you—usually your money, personal information, or passwords. The person who does this is called a **scammer**.

- **Phishing** (said like “fishing”): A common type of scam where scammers pretend to be a trusted person or company (like your bank, Netflix, or even a friend) to “fish” for your private details. They send fake messages hoping you’ll “bite” and give away your information.

2. Types of Online Scams

- **Fake Prize or Lottery Scams:** “Congratulations! You’ve won an iPhone 15! Click here to claim your prize!” You never entered any contest, so this is a lie to get you to click a bad link or pay a “fee” to get a prize that doesn’t exist.
- **Impersonation Scams:** Someone creates a fake profile pretending to be your friend, family member, or even a government official. They might send a message like, “Hi, it’s your cousin David. I’m in trouble, send me money urgently.”
- **Tech Support Scams:** A scary pop-up appears on your screen saying, “VIRUS DETECTED! Call this number NOW for help!” This is fake. Real companies don’t send warnings like this. They want you to call so they can trick you into paying for fake services or gain access to your computer.



3. Recognizing Scam Tactics Scammers use tricks to make you act quickly without thinking:

- **Too Good to Be True:** Offers for free money, expensive gifts (phones, cars), or unbelievable deals. If it seems too good, it probably is false.
- **Urgency and Fear:** They create panic: “Your account will be closed in 1 hour!” or “Your brother is in the hospital, send money now!” They don’t want you to have time to think or ask an adult.
- **Fake Authority:** Using logos of real companies (like GTBank, DSTV), fake email addresses that look almost real (e.g., support@gtbannk.com instead of support@gtbank.com), or pretending to be a police officer.

4. Protecting Yourself – The “Stop, Look, Think” Rule

- **STOP:** Don’t click, don’t reply, don’t panic.
- **LOOK:** Look carefully at the message. Check the sender’s email address. Is there strange spelling? Are they using urgency or fear?

- **THINK:** “Does this make sense?” “Would this company really contact me this way?” “Have I verified this with someone I trust?”
- **ACT:** **Never** give out personal information (address, parent’s bank details). **Do not** click on links or download attachments. **Tell** a parent, guardian, or teacher immediately.

EVALUATION

1. What is the main goal of a scammer?
2. A message says you won a free laptop for being the 1 millionth visitor to a website. What should you do?
3. Which feeling are scammers trying to create when they say “ACT NOW OR YOU WILL LOSE!”?
4. What is a “phishing” scam trying to steal from you?
5. If you get a strange message from a friend asking for money, what is the safest first step?
6. True or False: A real bank will email you and ask for your account password and PIN.

ASSIGNMENT

1. **Written:** List two “red flags” or warning signs that a message might be a scam.
2. **Creative:** Design a small poster with the slogan “**Stop, Look, Think Before You Click!**” Use colors and simple drawings.
3. **Oral:** Ask a parent or older sibling if they have ever received a scam call or message. Report back one example they shared (without sharing private details).

WEEK 3: DIGITAL FOOTPRINT

OBJECTIVES By the end of the lesson, pupils should be able to:

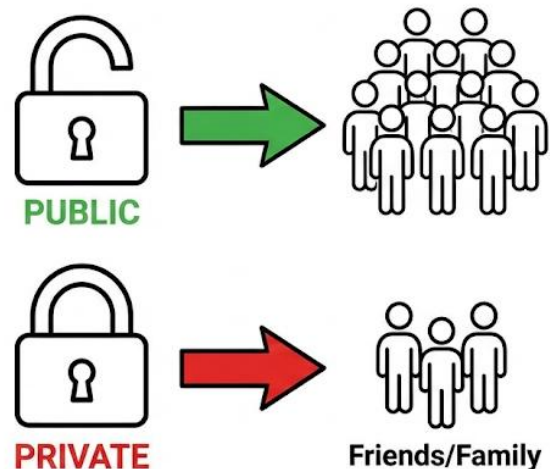
1. Define Digital Footprint.
2. Explain the permanent nature of online posts.
3. Mention two ways to manage a good reputation online.
4. Identify appropriate privacy settings.

CONTENT

1. Understanding Digital Footprint Your **Digital Footprint** is the trail or record of **everything you do online**. It's like invisible footprints you leave behind in the digital sand every time you use the internet.



DIGITAL FOOTPRINT



PRIVACY SETTINGS

- **What creates it?** The websites you visit, the videos you watch on YouTube, the photos and comments you post on social media or in class groups, the games you play online, and the things you search for.
- **Active Footprint:** Things you intentionally put online (posting a photo, commenting).

- **Passive Footprint:** Information collected about you without you actively posting it (websites tracking what you look at).

2. The Internet Never Forgets A common mistake is thinking “If I delete it, it’s gone forever.”

This is **not true**.

- **Permanent:** Once you post a photo or comment, it can be **saved** (screenshot), **shared**, or **reposted** by others in seconds. Even if you delete it from your profile, copies might still exist on other people’s devices or internet servers.
- **Think Before You Post (The Grandma Rule):** Before you post anything, ask yourself: “Would I be happy if my grandma, my teacher, or my future boss saw this?” If the answer is NO, don’t post it.

3. Being a Responsible Digital Citizen Your digital footprint shapes your **online reputation**—what people think of you based on what they see online.

- **Be Kind and Respectful:** Do not post mean comments, bully others, or share embarrassing things about people. This is called **cyberbullying** and it hurts people deeply.
- **Be Positive and Helpful:** Share good ideas, compliment others’ work, and post things that are truthful and helpful.
- **Future Impact:** Universities and employers sometimes look at applicants’ online profiles. A negative digital footprint with rude comments or inappropriate pictures can hurt your chances in the future.

4. Managing Your Privacy You can control who sees your information using **Privacy Settings**.

- **Public:** Anyone on the internet can see what you post.
- **Private/Friends Only:** Only people you approve as “friends” or “followers” can see your posts.
- **Advice for Safety:** For pupils, it is safest to keep social media accounts (if you have any) on **private** and only connect with people you know **in real life**. Always ask a parent to help you check these settings.

EVALUATION

1. What is a “digital footprint”?
2. If you delete a rude comment you posted yesterday, can you be sure no one ever saw it?
3. Why is the “Grandma Rule” a good guide for posting online?
4. What is one negative thing that could happen in the future because of a bad digital footprint?
5. If your social media account is set to “Public,” who can see your posts?
6. What should you do if someone is posting mean comments about a classmate online?

ASSIGNMENT

1. **Drawing:** Draw your own “Digital Footprint.” Inside a large footprint shape, write or draw 3 things that might be in your footprint (e.g., “Maths game scores,” “Google searches for projects,” “Photos with family”).
2. **Written:** Write one sentence explaining why posting a mean comment online is worse than saying it and apologizing immediately.
3. **Practical (with permission):** With a parent, check the privacy settings on one app you use (like YouTube or a game). Report if it is mostly set to public or private.

WEEK 4: COPYRIGHT AND DIGITAL ETHICS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Copyright.
2. Explain why we must respect others' work.
3. Define Plagiarism.
4. Mention how to use materials fairly for education.

CONTENT

1. Understanding Copyright Copyright is a law that protects the work people create. It means the **creator owns their work**. Just because you can find a picture, song, or article online doesn't mean you are free to use it however you want.

- **Symbol:** The copyright symbol is ©.
- **What is protected?** Books, music, movies, paintings, photographs, software, and even websites.
- **It means:** You cannot **copy, sell, change, or repost** someone else's work as your own without their permission.

2. Respecting Others' Work Online Finding an image on **Google** does **not** mean it is free to use. Google is a search engine, not a owner of the pictures.

- **Permission:** To use someone's work (e.g., a photographer's picture in your project), you should ask them for permission.
- **Creative Commons:** Some creators use **Creative Commons (CC)** licenses. This means they allow others to use their work for free, usually if you **give them credit** (say who made it). Look for symbols like CC BY.



3. Plagiarism Awareness

Plagiarism is stealing someone else's work or ideas and presenting them as your own. It is cheating.

- **Example:** Copying a paragraph from Wikipedia for your history homework and not saying where it came from.
- **Consequence:** In school, plagiarism can lead to a **zero mark**, a bad report, or other discipline. It is dishonest.
- **It's not just words:** Copying a friend's drawing or project idea without credit is also a form of plagiarism.

4. Citing Sources and Fair Use To avoid plagiarism, you must **cite your sources**. This means giving credit to the original creator.

- **How to cite simply:** Write where you got the information. Example: "According to the National Geographic Kids website, lions sleep for 20 hours a day."
- **Fair Use:** This is a special rule that allows you to use **small parts** of copyrighted work for schoolwork, news, or research **without** permission, as long as you **give credit**. You

can quote a few lines from a book for a book report, but you cannot photocopy the whole book.

A simple two-column table:

Do (Fair Use)	Don't (Plagiarism)
Use a small quote and say who wrote it.	Copy a whole page from a website.
Tell the artist's name when using their CC image.	Save a cool picture and pretend you drew it.
Write your own sentences about what you learned.	Let your older sibling do your homework for you.

EVALUATION

1. What does the © symbol tell you about a book or song?
2. What is plagiarism?
3. Is it okay to copy a chart from a website for your project if you change the colors?
4. What should you do if you want to use a “Creative Commons” photo in your presentation?
5. What does “citing a source” mean?
6. Who owns the story you wrote and drew pictures for in English class?

ASSIGNMENT

1. **Written:** In your own words, write a one-sentence definition of “Plagiarism.”
2. **Practical:** Look at the first few pages of any textbook. Find the copyright symbol © and write down the **year of publication** and the **publisher's name**.
3. **Creative:** Draw a poster for your class that says: “Be Honest Online: Give Credit for Other People's Work!”

WEEK 5 & 6: MIDTERM EXAMINATION AND BREAK



WEEK 7: BLOCK CODING WITH SCRATCH I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Review the Scratch interface.
2. Use motion blocks to make a sprite glide and bounce.
3. Change costumes to create animation.
4. Add a backdrop to a project.

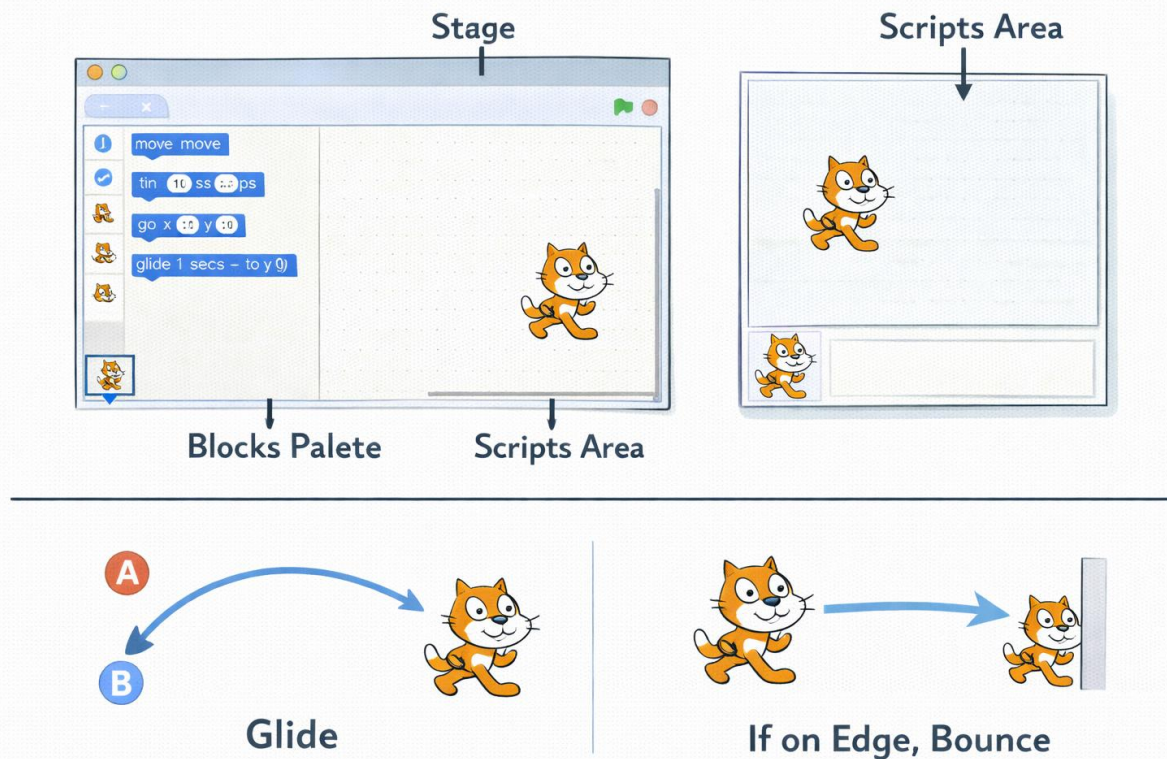
CONTENT

1. Review of Scratch Basics Scratch is a visual programming language where you drag and connect colorful code blocks to create stories, games, and animations.

- **Sprite:** Any character or object in your project (like the default cat, or a ball, a person). You can have many sprites.
- **Stage:** The background area where your sprites perform and move.
- **Blocks Palette:** The area on the left with color-coded blocks (Motion-blue, Looks-purple, Sound-pink, etc.).
- **Scripts Area/Coding Area:** The large middle area where you snap blocks together to create scripts (programs) for your sprites.

2. Advanced Motion Blocks Let's make sprites move in more interesting ways!

- **Glide Block:** glide (1) secs to x: (0) y: (0) makes the sprite slide smoothly from its current position to a new point on the stage. You can change the time (seconds) and the X (horizontal) and Y (vertical) coordinates.
- **If on Edge, Bounce:** This block makes a sprite turn around and move in the opposite direction when it touches the edge of the stage. This keeps it from disappearing off the screen!
- **Point Towards:** You can make a sprite always point towards another sprite or the mouse-pointer, which is useful for aiming.



3. Costume Changes (Animation) Animation is an illusion of movement created by showing a series of pictures quickly.

- **Costumes:** These are different pictures or “poses” for a single sprite. A walking cat sprite might have two costumes: one with left paw forward, one with right paw forward.
- **Creating Animation:** Use a forever loop with a next costume block inside. Also, add a wait (0.2) seconds block to control the speed. This makes the sprite switch between its costumes, making it look like it’s walking, talking, or changing expression.

4. Using Backdrops The **Backdrop** is the background picture on the stage. You can have multiple backdrops to change the scene.

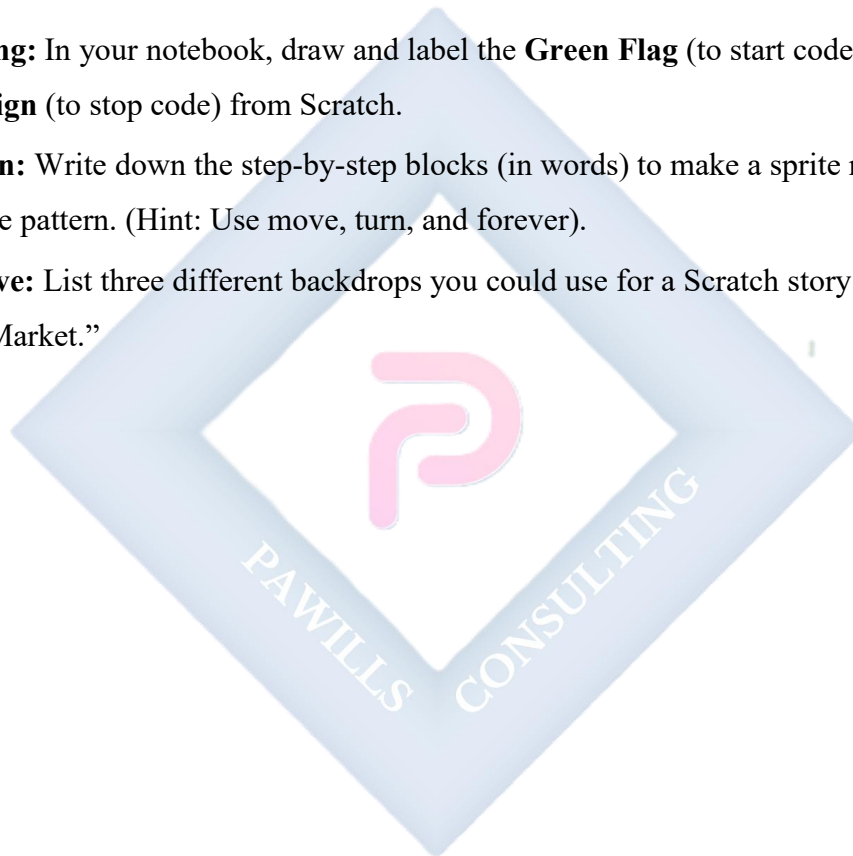
- **Adding:** Choose a backdrop from the library (like “Space,” “School,” “Forest”) or paint your own.
- **Changing with Code:** Use the purple Looks block switch backdrop to [] to change the background during your story or game (e.g., when the sprite goes through a door).

EVALUATION

1. What is the name of the character you program in Scratch?
2. Which block would you use to make a sprite slide smoothly to the center of the stage?
3. Why is the if on edge, bounce block useful in a game?
4. How do you make a sprite look like it is walking?
5. What is the difference between a “Sprite” and a “Backdrop”?
6. Which block makes other blocks inside it repeat over and over again forever?

ASSIGNMENT

1. **Drawing:** In your notebook, draw and label the **Green Flag** (to start code) and the **Red Stop Sign** (to stop code) from Scratch.
2. **Written:** Write down the step-by-step blocks (in words) to make a sprite move forever in a square pattern. (Hint: Use move, turn, and forever).
3. **Creative:** List three different backdrops you could use for a Scratch story about “A Day at the Market.”



WEEK 8: BLOCK CODING WITH SCRATCH II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a Variable.
2. Use Operator blocks for math.
3. Create a simple timer or scorekeeper.
4. Use the “Ask and Wait” block for user input.

CONTENT

1. Understanding Variables

A **Variable** is like a **labeled box or container** where you can store information that can **change** (vary) while your program runs.

- **Real-life Example:** The scoreboard in a football match. The “Score” starts at 0. When a team scores, the number in the “Score” box changes to 1, then 2, etc.
- **In Scratch:** You create a variable (like “Score,” “Lives,” “Timer,” “HighScore”) in the orange Variables category. Once created, you can set, change, and show it.

2. Using Operators Operators (green blocks) are used for **Math** and making **Decisions** by comparing things.

- **Math Operators:** + (add), - (subtract), * (multiply), / (divide). Example: (5 + 3) gives 8.
- **Comparison Operators:** < (less than), > (greater than), = (equal to). These are used inside if blocks to check conditions. Example: if <Score > 10> means “if the Score is greater than 10.”

3. Creating a Scorekeeper

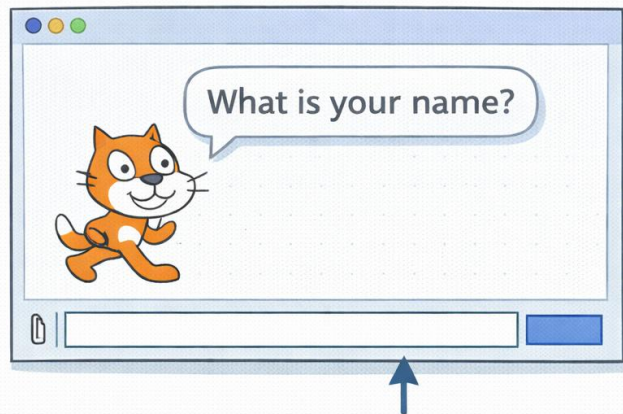
Let’s make a simple game score system:

1. Go to the Variables category and click “Make a Variable.” Name it “**Score.**”
2. When the green flag is clicked, set [Score v] to [0] to start the score at zero.
3. Whenever the player does something good (e.g., clicks a sprite, catches an object), use change [Score v] by [1] to add a point.

4. Make sure to show the variable so you can see it on the stage.

4. User Input (Asking Questions) You can make your program interactive by asking the user (player) for information.

- Use the light blue Sensing block: ask [What is your name?] and wait.
- When this block runs, a small box appears at the bottom of the stage for the player to type an answer.
- The answer is stored in a special variable called answer. You can use it to make the sprite say, join [Hello] (answer).



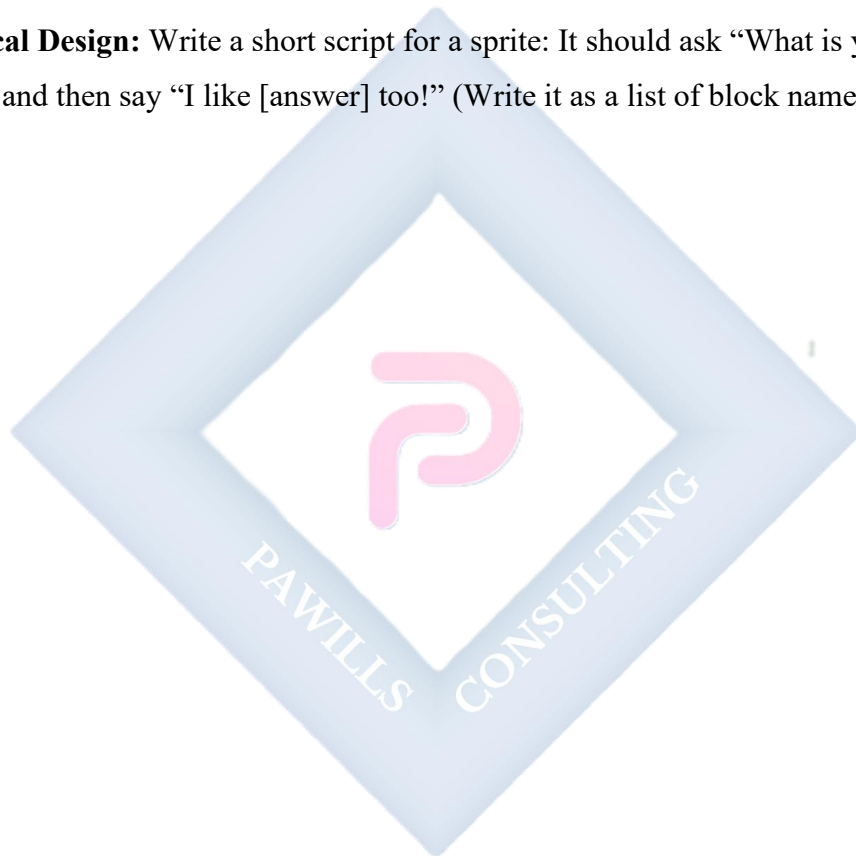
EVALUATION

1. What is a variable in programming? Give one example.
2. Which color are the Operator blocks in Scratch?
3. In a catching game, what block would you use to increase the score by 1 point each time an apple is caught?
4. How do you start a "Score" variable at zero when the game begins?

5. Which block allows the player to type their name into your Scratch project?
6. What symbol is used to check if two values are the same (e.g., if Score equals 5)?

ASSIGNMENT

1. **Written:** Imagine you are creating a game. Create a variable named “Health” on paper. Write the code (in words) to: “Set Health to 100 when the game starts, and change Health by -10 if the sprite touches an enemy.”
2. **Logical:** If Score = 7 and Target = 10, is Score > Target true or false?
3. **Practical Design:** Write a short script for a sprite: It should ask “What is your favorite food?” and then say “I like [answer] too!” (Write it as a list of block names).



WEEK 9: PROGRAMMING A GAME ENVIRONMENT

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List the three basic parts of a game design.
2. Create and control a game sprite.
3. Explain Collision Detection.
4. Test a simple game.

CONTENT

1. Game Design Basics

Before coding, every good game needs a plan with these three parts:

- **Objective/Goal:** What is the player trying to do? (e.g., Collect 10 coins, Reach the finish line, Avoid enemies for 60 seconds).
- **Rules:** What can the player do? What can't they do? (e.g., You can jump over pits. You cannot touch the fire. You have 3 lives).
- **Characters & Objects:** Who is the **player** (hero)? Who or what are the **obstacles** (enemies, traps)? What are the **rewards** (coins, power-ups)?

2. Movement Controls To play, the user must control the main character (sprite). We use the Events (yellow) blocks with keyboard keys.

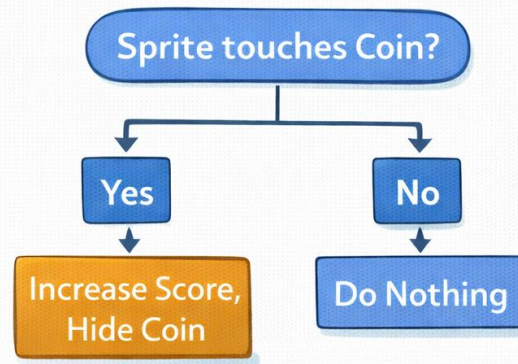
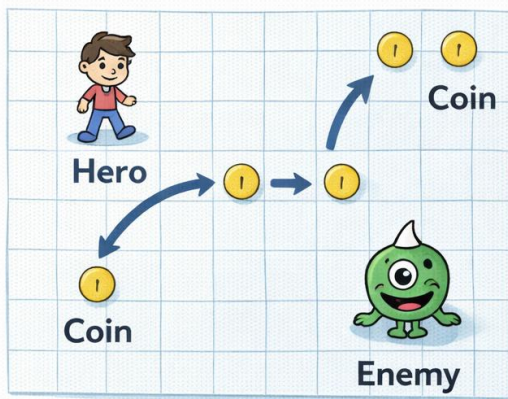
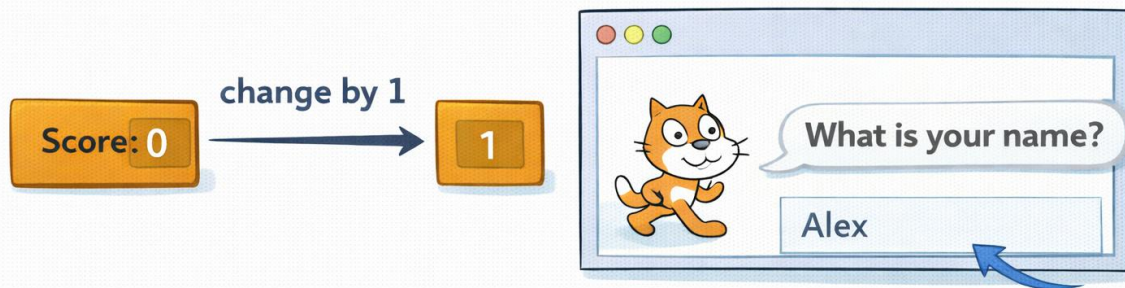
- **Arrow Key Controls:**
 - When [right arrow v] key pressed → change x by (10) (Moves Right)
 - When [left arrow v] key pressed → change x by (-10) (Moves Left)
 - When [up arrow v] key pressed → change y by (10) (Moves Up)
 - When [down arrow v] key pressed → change y by (-10) (Moves Down)
- **X and Y:** Remember, the stage is a grid. **X** controls left/right position. **Y** controls up/down position.

3. Collision Detection This is how the computer knows when two sprites **touch** each other. It is a crucial part of game rules.

- **How to Use It:** Use the light blue Sensing block touching []? inside a Control block if <> then.
- **Example 1 (Lose Life):** if <touching [Enemy v] ?> then → change [Lives v] by (-1)
- **Example 2 (Collect Reward):** if <touching [Coin v] ?> then → change [Score v] by (1)
→ hide (to make the coin disappear)

4. Game Testing Testing means playing your own game to find and fix problems (bugs).

- **Ask yourself:** Are the controls smooth? Is the game too easy or too hard? Do the rules work correctly (e.g., does the score really go up when you touch a coin)?
- **Get Feedback:** Let a friend play. Watch where they get confused or what they enjoy. Use their feedback to improve your game.



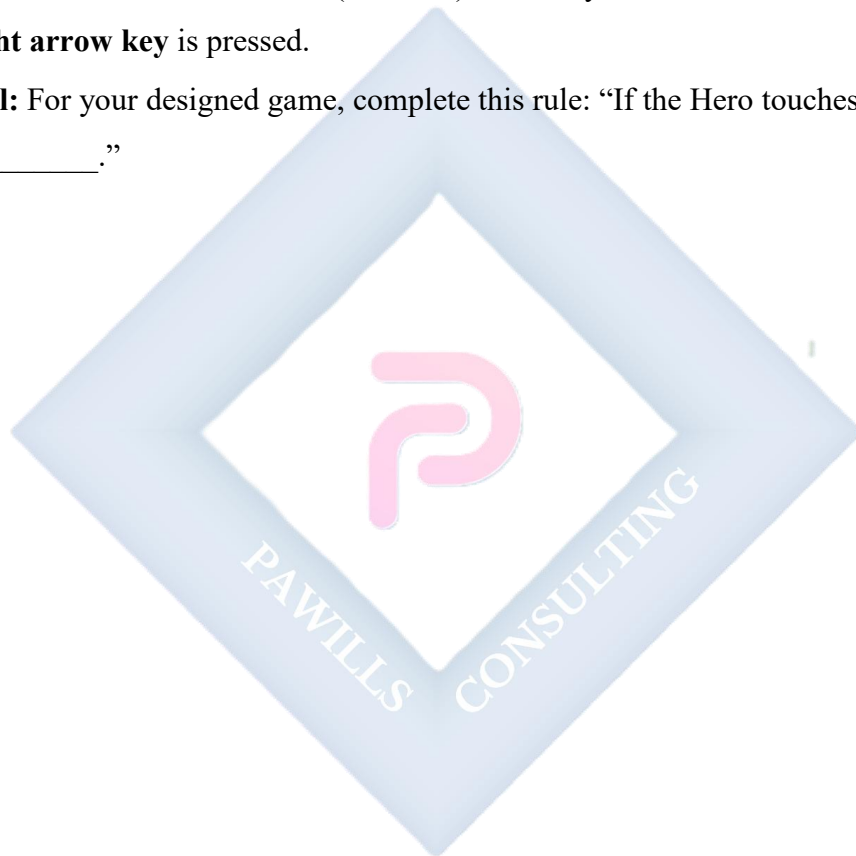
EVALUATION

1. What is the “objective” in a game of catching falling fruit?
2. Which coordinate (X or Y) do you change to make a sprite move upwards?
3. What is “collision detection”?

4. Which code block checks if the player sprite is touching an enemy?
5. If your player moves left when you press the right arrow, what kind of error is that?
6. Why is it important to let someone else test your game?

ASSIGNMENT

1. **Design:** On paper, design a simple game. Draw the **Hero**, one **Enemy**, and the **Goal** (e.g., a door or a treasure chest).
2. **Written:** Write the Scratch code (in words) to make your Hero move to the **right** when the **right arrow key** is pressed.
3. **Logical:** For your designed game, complete this rule: “If the Hero touches the Enemy, then _____.”



WEEK 10: DEBUGGING AND PROBLEM SOLVING

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a Bug and Debugging.
2. Identify syntax and logic errors.
3. Mention two strategies for debugging.
4. Explain problem decomposition.

CONTENT

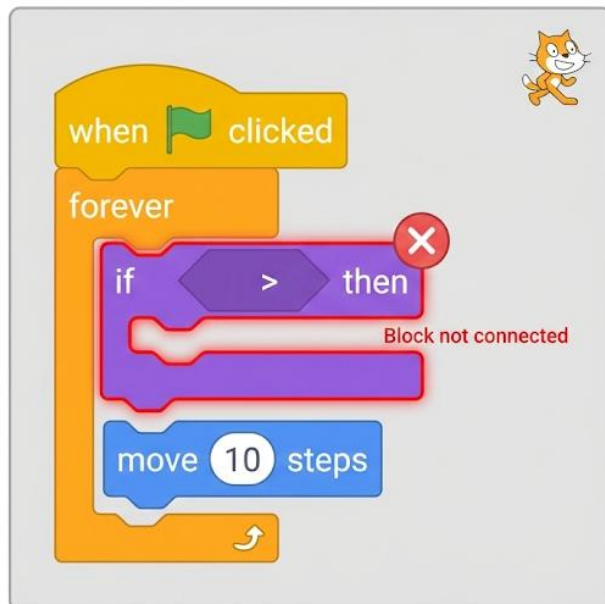
1. What is a Bug? In computer programming, a **bug** is a mistake or error in the code that causes the program to behave in an unexpected or wrong way. The process of finding and fixing these mistakes is called **debugging**.

- **Fun Fact:** The term “bug” was used when a real moth got stuck in an old computer and caused a problem!

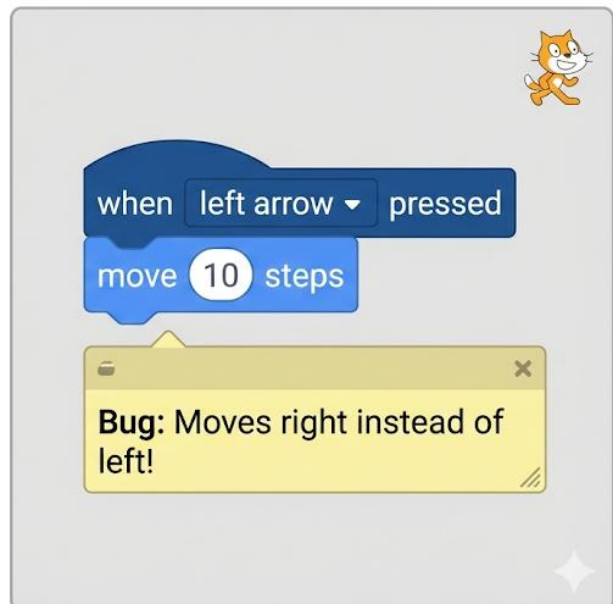
2. Types of Errors

- **Syntax Error:** This is like a **spelling or grammar mistake** in your code. The computer doesn't understand the instruction. In Scratch, blocks might not snap together, or you might leave a required field empty. The computer says, “I don't understand what you want me to do.”
- **Logic Error:** This is trickier. The code runs without crashing, but it does **the wrong thing**. This is because the programmer's logic (plan) was flawed. Example: You want the sprite to move right, but you used change x by (-10), so it moves left instead. The computer is doing exactly what you told it, but what you told it was incorrect.

SYNTAX ERROR



LOGIC ERROR



3. Debugging Strategies How to be a code detective:

1. **Read Your Code Aloud:** Explain each block step-by-step to yourself or a friend. Often, you'll hear the mistake as you say it.
2. **Test in Small Pieces:** Don't write all your code and then run it. Test small sections as you build. Click on a stack of blocks to run just that part and see if it works.
3. **Check Your Assumptions:** Are you sure the "up arrow" block is connected? Did you set the variable correctly? Go back and double-check the basics.
4. **Take a Break:** Sometimes, walking away from the problem for a few minutes helps you see it with fresh eyes.

4. Decomposition This is a big problem-solving strategy. **Decomposition** means breaking a big, complicated problem down into smaller, simpler, more manageable parts.

- **Example – Big Problem:** "Create a racing game."
- **Decomposed Steps:**
 - a. Make a car sprite.
 - b. Make the car move with arrow keys.
 - c. Draw a road backdrop.
 - d. Add another car as an obstacle.

- e. Detect collision with the other car (game over).
- f. Add a finish line and a timer. Now, you can solve **one small step at a time** instead of being overwhelmed.

EVALUATION

1. What is “debugging”?
2. If your code has a missing piece and won’t run at all, what type of error is it most likely to be?
3. Give an example of a logic error in a Scratch game.
4. What does “decomposition” mean in problem-solving?
5. Why is it helpful to test your code after adding just 2 or 3 new blocks?
6. If your sprite is supposed to jump when you press the spacebar but nothing happens, what is the first thing you should check?

ASSIGNMENT

1. **Practical (Unplugged):** Decompose the task “Making a Plate of Fried Rice” into 5 smaller steps. Write them in order.
2. **Drawing:** Draw a magnifying glass. Around it, write the words: “FIND THE BUG!”
3. **Written:** Look at this buggy instruction: “To tie your shoe: 1. Pull laces tight. 2. Put on shoe. 3. Make a bow.” What is the logic error? Write the correct order.

WEEK 11: REVISION

CONTENT

1. Review of Term 2 Topics

- **Digital Citizenship:** Internet Safety Rules, Recognizing Scams, Digital Footprint, Copyright/Plagiarism.
- **Coding & Game Design:** Scratch Interface, Motion & Animation, Variables & Operators, User Input, Game Controls, Collision Detection, Debugging.

2. Practical Project – “Catch the Falling Objects” Game Pupils will apply their skills to finish a simple game project in Scratch based on a template or their own Week 9 design.

- **Objective:** Catch as many falling fruits (good objects) as possible while avoiding rocks (bad objects).
- **Requirements:**
 - A controllable basket (sprite).
 - Falling fruit and rock sprites.
 - A “Score” variable that increases for fruit.
 - A “Lives” variable that decreases for rocks.
 - Game-over message when lives reach zero.
- **Presentation & Feedback:** Pupils will present their finished game to the class. Classmates will play and give one piece of positive feedback and one suggestion.

EVALUATION

- Informal assessment through observation during the practical project work.
- Q&A session focusing on key vocabulary and concepts from the term (e.g., “What is phishing?”, “What does a variable do?”).

ASSIGNMENT

1. **Study Skills:** Review your notes from Weeks 1-10. Create 5 flash cards with a term on one side (e.g., “Digital Footprint”) and the definition on the other.

2. **Preparation:** Write down one question about a topic from this term that you still find a little confusing, to ask your teacher next week.
3. **Practical:** Ensure your Scratch game project is saved with a clear name:
YourName_CatchGame.



PRIMARY 6 BASIC DIGITAL LITERACY SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Advanced Robotics Concepts	Robotic sensors: ultrasonic, infrared, touch, light, temperature. Actuators: motors, servos. Robot programming logic. Obstacle avoidance. Line following basics. Robot competitions overview.
2	Physical Computing	Introduction to microcontrollers: Arduino basics. Input and output pins. LED control. Button inputs. Simple circuits. Safety with electronics. Real-world physical computing applications.
3	Machine Learning Basics	Understanding machine learning: computers learning from data. Supervised vs unsupervised learning. Training models with examples. Image recognition, text classification. Bias in AI. Practical ML demonstrations.
4	AI in Pictoblox	Pictoblox AI extensions. Image recognition projects. Speech recognition. Text-to-speech. Pose detection. Creating AI-powered games. Combining AI with robotics simulations.
5	MIDTERM EXAMINATION	Assessment covering Weeks 1-4
6	MIDTERM BREAK	Holiday
7	Game Development Project I	Complete game development cycle. Game design document creation. Choosing game genre. Designing characters and levels. Planning game mechanics. Storyboarding. Asset creation or sourcing.
8	Game Development Project II	Implementing game in Scratch or Pictoblox. Programming game logic. Adding multiple levels. Score systems and lives. Game testing and debugging. Sound effects and music.

9	Technology and Society	Technology impact on daily life. Digital divide: access inequality. Technology careers overview. Ethical technology use. Environmental impact of technology. E-waste awareness. Sustainable technology practices.
10	Preparing for Digital Future (	Future technology trends: VR, AR, IoT, quantum computing. Skills for future: coding, creativity, critical thinking, collaboration. Continuous learning importance. Building digital portfolio. Resources for further learning. Transition to secondary school ICT.
11	REVISION	Comprehensive review of entire primary school digital literacy curriculum. Final capstone projects combining multiple skills. Presentations and demonstrations.
12	EXAMINATION	End of Third Term/Annual Examination and Primary School Final Examination (Theory and Practical)
13	CLOSING/GRADUATION	Report distribution, awards ceremony, graduation to secondary school

PRIMARY 5 BASIC DIGITAL LITERACY LESSON NOTES

THIRD TERM

WEEK 1: SEQUENCE IN PROGRAMMING

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define sequence in programming.
2. Explain the importance of order in code.
3. Plan a simple sequence using step-by-step instructions.
4. Create a multi-step program on paper or a computer.

CONTENT

1. Deep Dive into Sequencing In programming, a **sequence** is a set of instructions given to a computer that must be followed **one after another, in a specific order**. Computers are not like humans; they cannot guess or skip steps. They follow commands exactly as given.

- **Real-World Example:** A recipe is a sequence. You must follow the steps in order to make the dish correctly. You can't bake a cake before mixing the ingredients.

2. Order Importance (The Sandwich Analogy) Think about making a peanut butter sandwich.

1. Take two slices of bread.
2. Spread peanut butter on one slice.
3. Put the slices together.
4. Cut the sandwich in half.

If you mix up the order—for example, if you try to cut the sandwich in half **before** you put the slices together—you will have a big mess! In coding, the **order of commands is just as important**. A wrong sequence creates a **bug** (an error).

3. Sequential Thinking Exercises (Algorithms) Before writing code, programmers plan the sequence of steps. This plan is called an **Algorithm**.

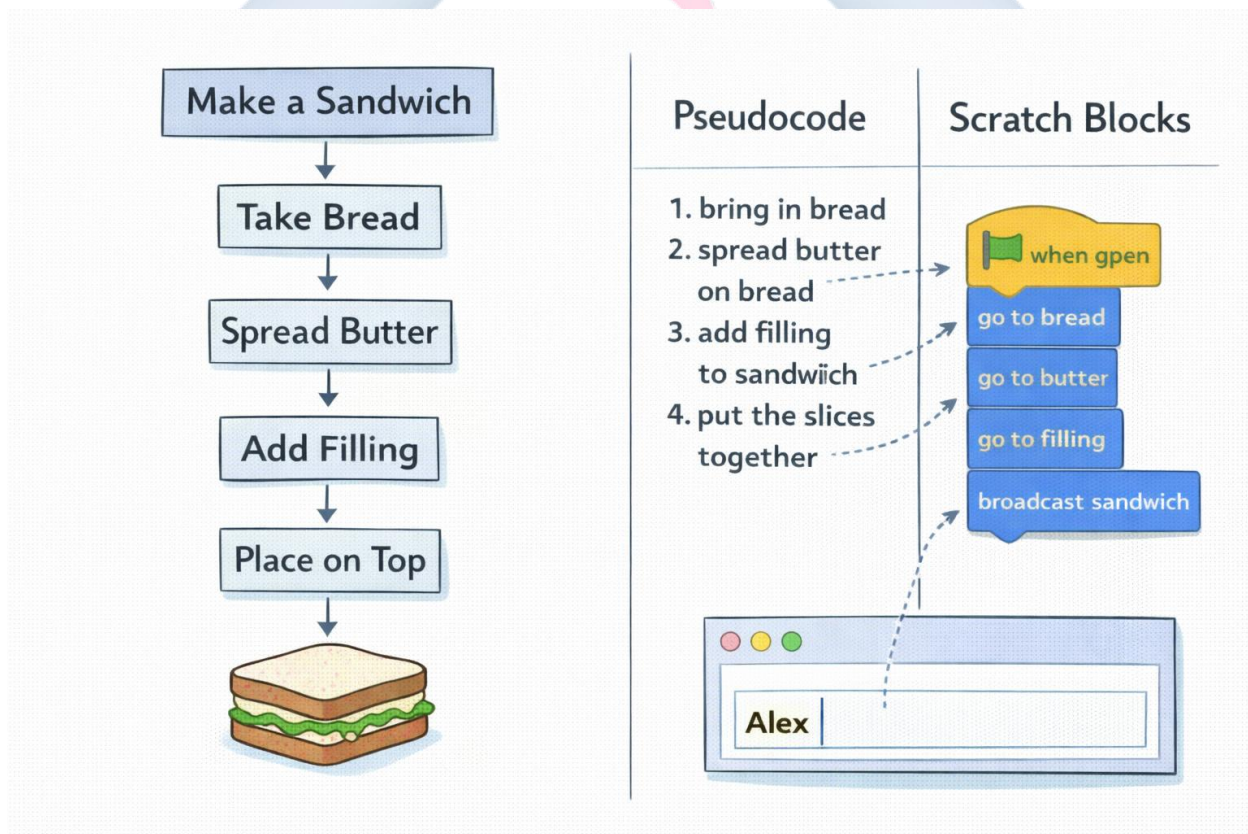
- **Algorithm:** A clear, step-by-step plan to solve a problem or complete a task.

- **Exercise:** Write an algorithm for “Brushing Your Teeth.”

- Pick up your toothbrush.
- Put toothpaste on the brush.
- Brush your teeth for two minutes.
- Spit out the toothpaste.
- Rinse your mouth and the brush.

4. Planning and Documenting Steps Good programmers **plan first, code later**. They use simple tools:

- **Pseudocode:** Writing the steps in plain English (or your own language), not in a programming language. Example: “Step 1: Make the cat sprite move 10 steps. Step 2: Make the cat say ‘Hello!’”
- **Flowchart:** A diagram that uses boxes and arrows to show the sequence of steps and decisions. A flowchart starts with an oval (Start) and ends with an oval (End).



EVALUATION

1. What do we call a set of instructions done in a specific order?
2. Why must a computer follow instructions in sequence?
3. What happens in a program if the sequence of steps is wrong?
4. What is the plain-English plan for a program called?
5. True or False: If you forget a step in your code, the computer can usually guess what you meant to do.

ASSIGNMENT

1. **Written:** Write a step-by-step algorithm (sequence) for the task: “Preparing to write in your exercise book.” (Hint: Steps might include: sit down, open bag, take out book, take out pen, open book to a new page).
2. **Drawing:** Draw a simple 4-step flowchart showing how to drink a cup of water. Use rectangles for actions and arrows to connect them.
3. **Logic:** Unscramble this sequence for turning on a computer: (a) Type your password, (b) Press the power button, (c) Wait for the home screen to load, (d) Sit at the desk. Write the letters in the correct order.

WEEK 2: LOOPS AND REPETITION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a loop in programming.
2. List three types of loops found in Scratch.
3. Explain the benefit of using loops.
4. Identify a nested loop.

CONTENT

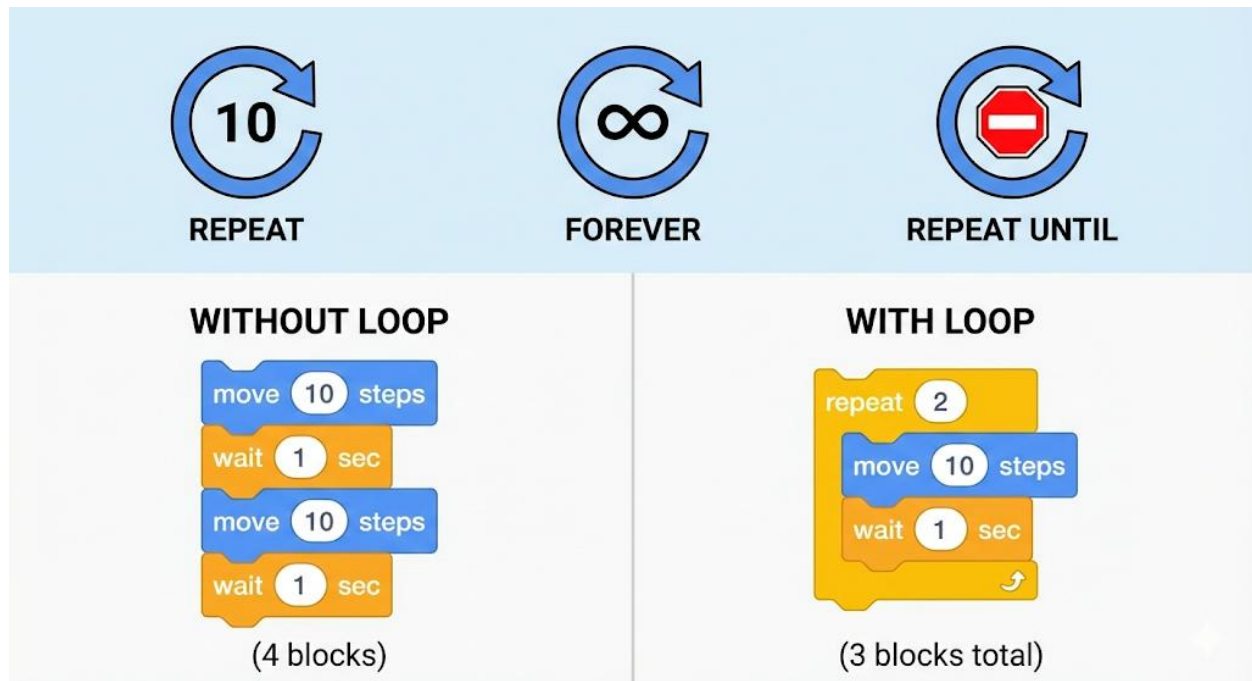
1. Understanding Loops

A **Loop** in programming is a command that tells the computer to repeat a set of instructions **multiple times**. Instead of writing the same code block over and over, you write it once and put it inside a loop. It's like telling a friend, "Clap your hands 10 times," instead of saying "Clap" ten separate times.

2. Types of Loops in Scratch

In Scratch, loops are found in the orange Control blocks.

- **Repeat (10) Loop:** This loop runs the code inside it a **specific number of times** you choose (e.g., 10 times) and then stops.
- **Forever Loop:** This loop runs the code inside it **endlessly**, until you press the stop button. It is perfect for games where actions need to continue, like having an enemy move constantly.
- **Repeat Until Loop:** This loop runs the code inside it **until a certain condition becomes true**. Example: Repeat until <touching [edge v]?> – the sprite will keep moving until it touches the edge of the stage.



3. Efficiency (Why Use Loops?) Loops make code **efficient**:

- **Shorter Code:** Your program uses fewer blocks, which makes it easier to read and fix.
- **Easier to Change:** If you want to repeat something 100 times instead of 10, you just change one number in the loop, instead of adding 90 more blocks!
- **Less Work:** You save time and effort.

4. Nested Loops A **Nested Loop** is a loop placed **inside another loop**. It's like a clock: The **second hand** makes a full loop 60 times for every **1 loop** of the **minute hand**.

- **Example:** You could have a repeat 4 loop (to draw a square) inside a repeat 10 loop (to draw 10 squares in a row). This creates complex patterns with simple code.

EVALUATION

1. What is the main job of a loop in programming?
2. Which loop would you use to make a sprite flash colors continuously during a game?
3. What is one major benefit of using a loop instead of copying and pasting the same code?
4. What is the color of the Loop blocks in Scratch?
5. What do we call a loop that is placed inside another loop?
6. If you put a say Hello! block inside a repeat 3 loop, how many times will the sprite speak?

ASSIGNMENT

1. **Drawing:** Draw the shape of the forever block and the repeat (10) block in your notebook.
2. **Written:** Write a short instruction using a loop: “Tell me how to jump 5 times.”
(Example: Repeat 5 times: Jump).
3. **Logical:** Look at this inefficient code: “Turn on light. Wait. Turn off light. Wait. Turn on light. Wait. Turn off light.” How could a loop make this better? Rewrite the instruction using a loop.



WEEK 3: CONDITIONALS (IF-THEN)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a conditional statement.
2. Identify “If-Then” and “If-Then-Else” blocks.
3. Use comparison operators (<, >, =).
4. Create a simple decision-making script.

CONTENT

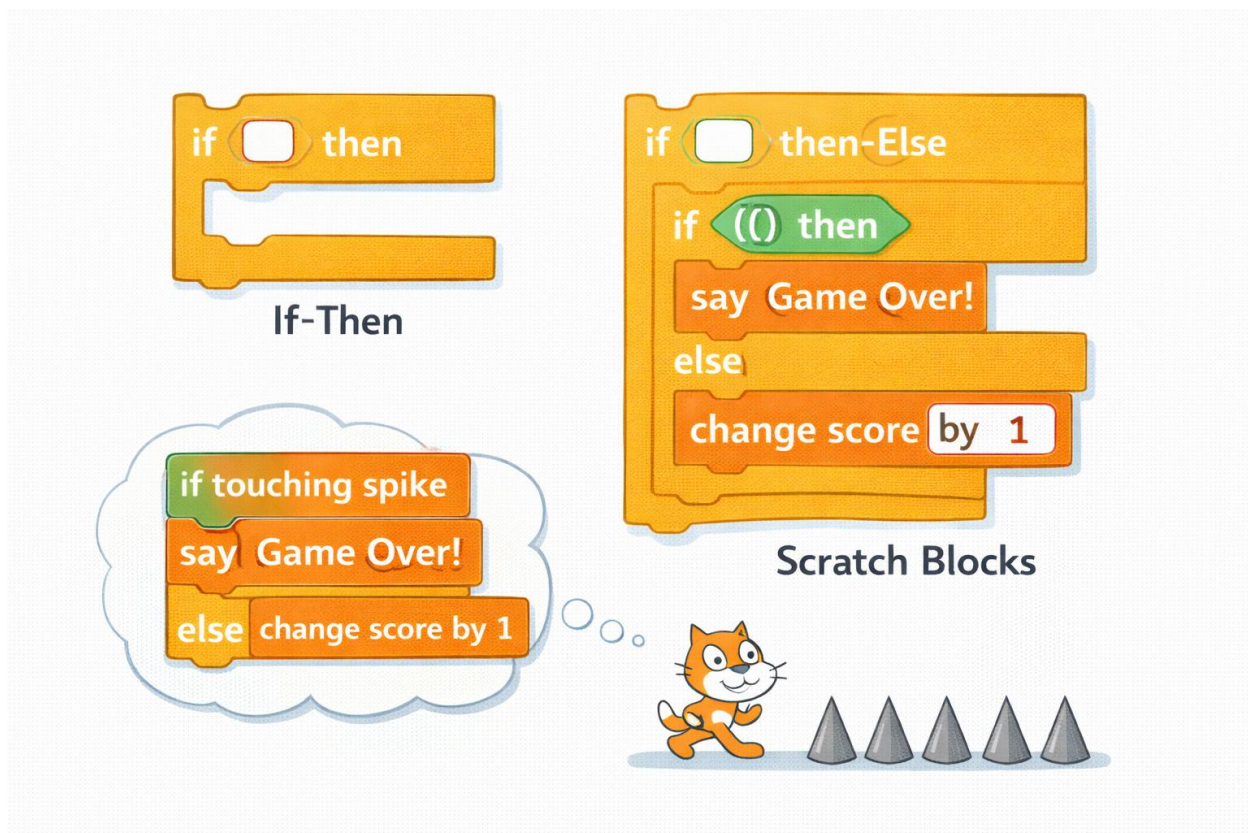
1. Understanding Conditions

A **Conditional Statement** is a command that lets a program **make decisions**. It checks if something is **true** or **false**, and then acts based on the answer. It gives your program a “brain” to choose between different paths.

- **Real-Life Example:** **IF** it is raining outside (condition is true), **THEN** take an umbrella. **ELSE** (condition is false, it is not raining), leave the umbrella at home.

2. Conditional Blocks in Scratch

- **If-Then Block:** Checks a condition. **If** the condition is **true**, it runs the code blocks inside it. **If** the condition is **false**, it skips them and does nothing.
- **If-Then-Else Block:** Checks a condition. **If** true, it runs the first set of blocks (the “Then” part). **If** false, it runs the second set of blocks (the “Else” part). This allows for **two different outcomes**.



3. Comparison Operators (The Green Blocks) To make decisions, computers need to compare values. We use green **Operator** blocks inside the condition's hexagon-shaped slot.

- **= (Equals):** Checks if two values are the same. If <Score = 10>
- **> (Greater Than):** Checks if the first value is larger. If <Time > 60>
- **< (Less Than):** Checks if the first value is smaller. If <Lives < 1>

4. Creating Responsive Programs Conditionals make programs interactive and smart.

- **Game Example:** If <touching [Enemy v] ?> then change [Lives v] by (-1) Else change [Score v] by (1)
- **Interactive Story Example:** If <key [space v] pressed?> then switch backdrop to [Castle v]

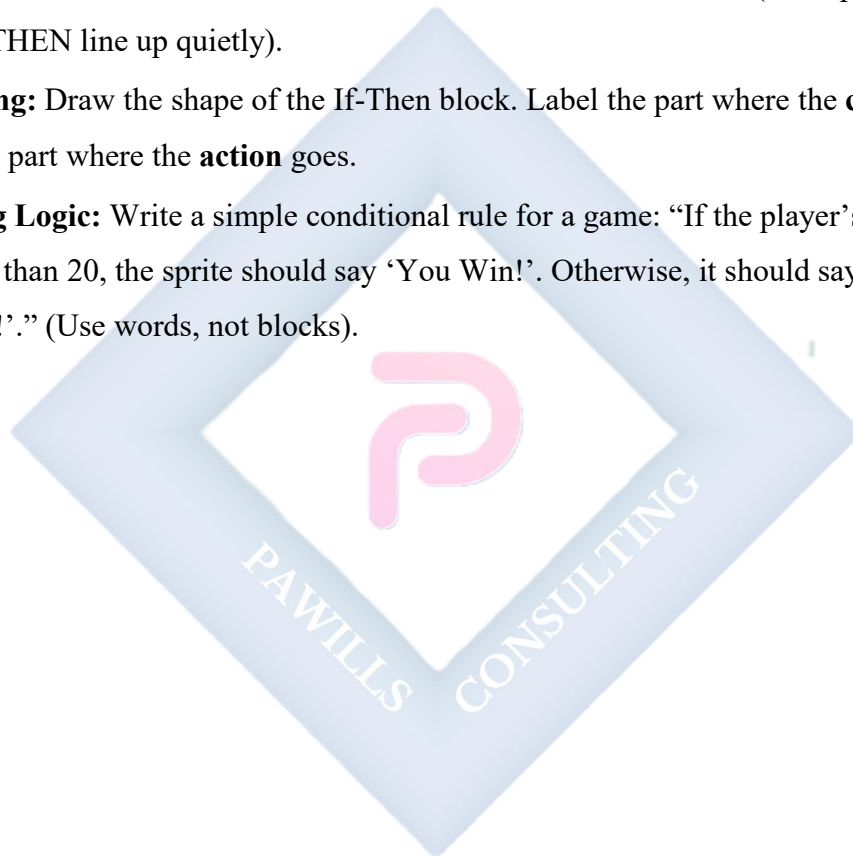
EVALUATION

1. What does a conditional statement allow a program to do?
2. Complete this real-life conditional: **IF** I am hungry, **THEN** _____.

3. Which conditional block should you use if you want one action to happen when true, and a *different* action to happen when false?
4. What color are the Operator blocks used for comparisons in Scratch?
5. Write the symbol for “Greater Than.”
6. Why is a conditional statement necessary for a game to know when a player wins or loses?

ASSIGNMENT

1. **Written:** Write three “If-Then” statements based on school rules. (Example: IF the bell rings, THEN line up quietly).
2. **Drawing:** Draw the shape of the If-Then block. Label the part where the **condition** goes and the part where the **action** goes.
3. **Coding Logic:** Write a simple conditional rule for a game: “If the player’s score is greater than 20, the sprite should say ‘You Win!’. Otherwise, it should say ‘Keep Trying!’.” (Use words, not blocks).



WEEK 4: BASICS OF ROBOTICS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a Robot.
2. List three types of robots.
3. Identify the main components of a robot (Sensors, Actuators, Processors).
4. Mention two uses of robots in daily life.

CONTENT

1. What is a Robot? A **Robot** is a **machine** that can be programmed (given code/instructions) by a computer to carry out tasks **automatically** (on its own). It is a combination of **hardware** (physical parts you can touch) and **software** (the code that controls it).

- **Key Idea:** Not all machines are robots. A robot is special because it can be programmed to do different things. A fan has one job; a robot can have many.

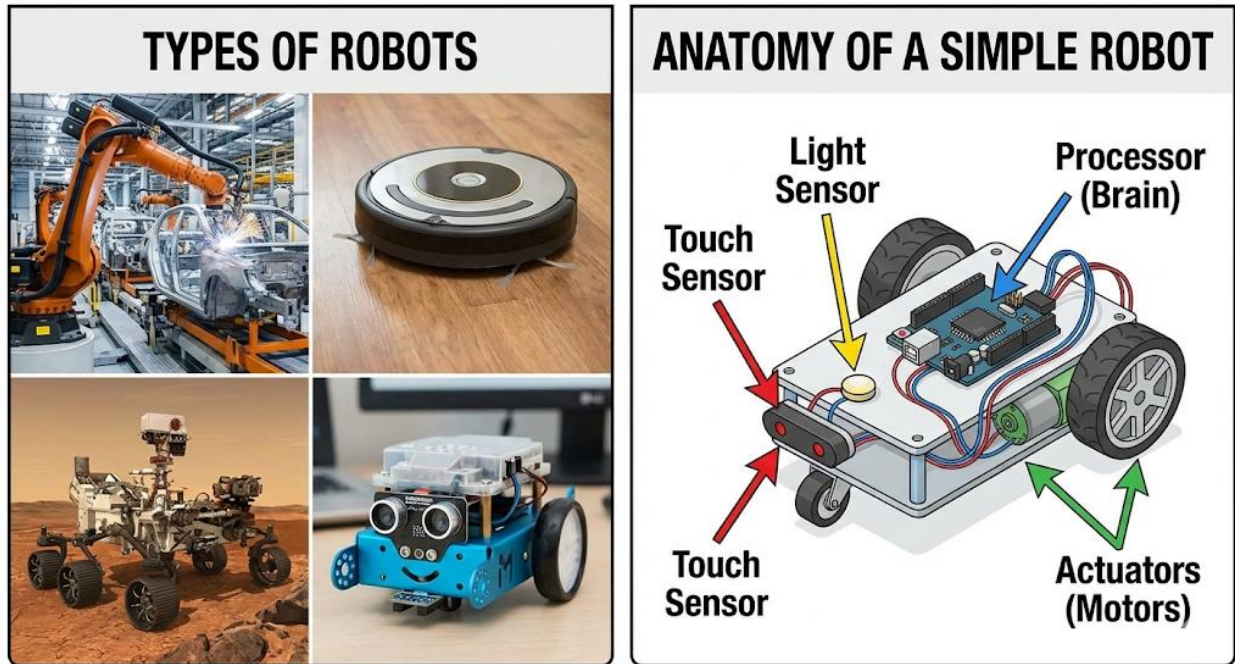
2. Types of Robots

- **Industrial Robots:** Large, strong arms in factories that weld car parts, paint, or assemble products very quickly and precisely.
- **Service Robots:** Help people with daily tasks. Examples: *Roomba* (vacuum cleaner robot), robots that help in hospitals to deliver medicine.
- **Exploration Robots:** Go to places too dangerous or far for humans. Examples: *Mars Rovers* (space), underwater robots exploring shipwrecks.
- **Educational Robots:** Used in schools for learning, like *mBots* or *Lego Mindstorms*. We build and code them ourselves!

3. Robot Components (Like Body Parts) All robots have three main parts that work together:

- **Sensors (The Senses):** These are the robot's **input** devices. They collect information from the world, just like our eyes, ears, and skin. Examples: Touch sensor, light sensor, ultrasonic sensor (like bat ears for distance), camera.

- **Processor (The Brain):** This is the **computer** or microcontroller (like a tiny CPU) inside the robot. It runs the program, reads data from the sensors, and makes decisions.
- **Actuators (The Muscles):** These are the **output** devices. They make the robot move or act based on the brain's decisions. Examples: Motors (to spin wheels or arms), speakers (to make sound), LED lights.



4. Robots in Daily Life Robots are already around us in Nigeria:

- **ATM Machines:** They are simple robots that follow your commands to give out cash.
- **Assembly Lines:** In factories making cars, drinks, or phones.
- **Drones:** Flying robots used for filming, delivering small packages, or surveying farmland.

EVALUATION

1. What makes a robot different from an ordinary machine like a blender?
2. Which part of a robot acts like its eyes and ears?
3. Which part of a robot is like its muscles, making it move?
4. Name one type of robot that works in a factory.
5. Give an example of a robot that might be used inside a home.
6. True or False: A robot can work without any program or code telling it what to do.

ASSIGNMENT

1. **Drawing:** Draw your own simple robot design. Label its **Sensor**, **Processor**, and **Actuator**.
2. **Creative:** If you had a personal robot at home, list three tasks you would program it to do for you.
3. **Written:** Write one sentence that explains the difference between a robot and a regular bicycle.



WEEK 5 & 6 MIDTERM EXAMINATION AND BREAK



WEEK 7: ROBOTICS APPLICATIONS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. List four fields where robots are used.
2. Explain how robots help humans in dangerous tasks.
3. Identify the “3 Ds” of robotics tasks.
4. Mention one educational robot kit.

CONTENT

1. Robots in Different Fields

Robots are useful tools in many important jobs:

- **Manufacturing & Factories:** Assembling cars, phones, and bottling drinks. They work fast, don't get tired, and make very few mistakes.
- **Healthcare:** Helping surgeons with very precise operations, delivering lab samples in hospitals, or helping patients with therapy.
- **Agriculture:** “Agrobots” can plant seeds, water crops, check soil health, and even pick fruits like strawberries.
- **Space & Deep Sea Exploration:** Sending robots like the Perseverance rover to Mars is safer and cheaper than sending humans. Underwater robots fix oil pipes or explore ocean depths.

2. How Robots Help Humans: The 3 Ds Robots are especially good at jobs that are difficult or unsafe for people. We call these the “3 Ds”:

- **Dull:** Jobs that are very repetitive and boring. Example: Putting the same screw into a product thousands of times a day.
- **Dirty:** Jobs in messy, unclean environments. Example: Cleaning inside sewers or chemical tanks.
- **Dangerous:** Jobs that could injure or harm a person. Example: Defusing bombs, fighting fires, working in radioactive areas, or deep mining.

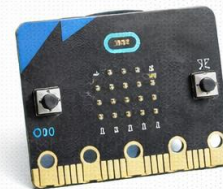
The 3 Ds		
		
Dull	Dirty	Dangerous



mBot



Lego Mindstorms



Micro:bit

3. Educational Robots (Learning Kits)

These are robots designed for students to learn coding and engineering.

- **Lego Mindstorms / Spike Prime:** You build robots using Lego bricks and then program them to move, sense, and complete challenges.
- **mBot:** A friendly, wheeled robot car that is easy to assemble and program using Scratch-like software.
- **Micro:bit:** A tiny programmable computer board that can be built into many different robot projects.

4. Robot Programming Basics Programming a robot often uses **block-based coding** (like Scratch) that we already know! Instead of making a cat move on screen, we send the code to the robot to make its real motors spin and its real sensors react to light or touch.

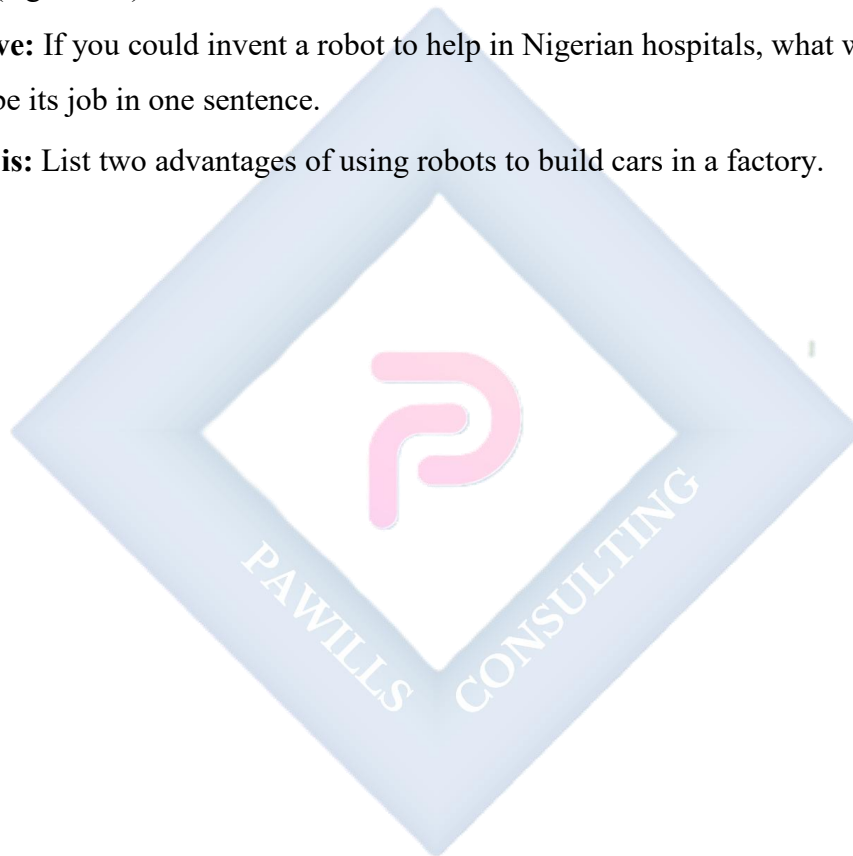
EVALUATION

1. How can robots help doctors in a hospital?
2. Why do scientists send robots to Mars instead of people?

3. What do the “3 Ds” stand for in robotics?
4. Give an example of a “Dull” job a robot could do in a factory.
5. Name one educational robot kit used in schools.
6. What is the main reason for using a robot in a “Dangerous” job?

ASSIGNMENT

1. **Research:** Find out and write the name of one famous robot that has explored another planet (e.g., Mars).
2. **Creative:** If you could invent a robot to help in Nigerian hospitals, what would it do? Describe its job in one sentence.
3. **Analysis:** List two advantages of using robots to build cars in a factory.



WEEK 8: GAME DEVELOPMENT WITH PICTOBLOX

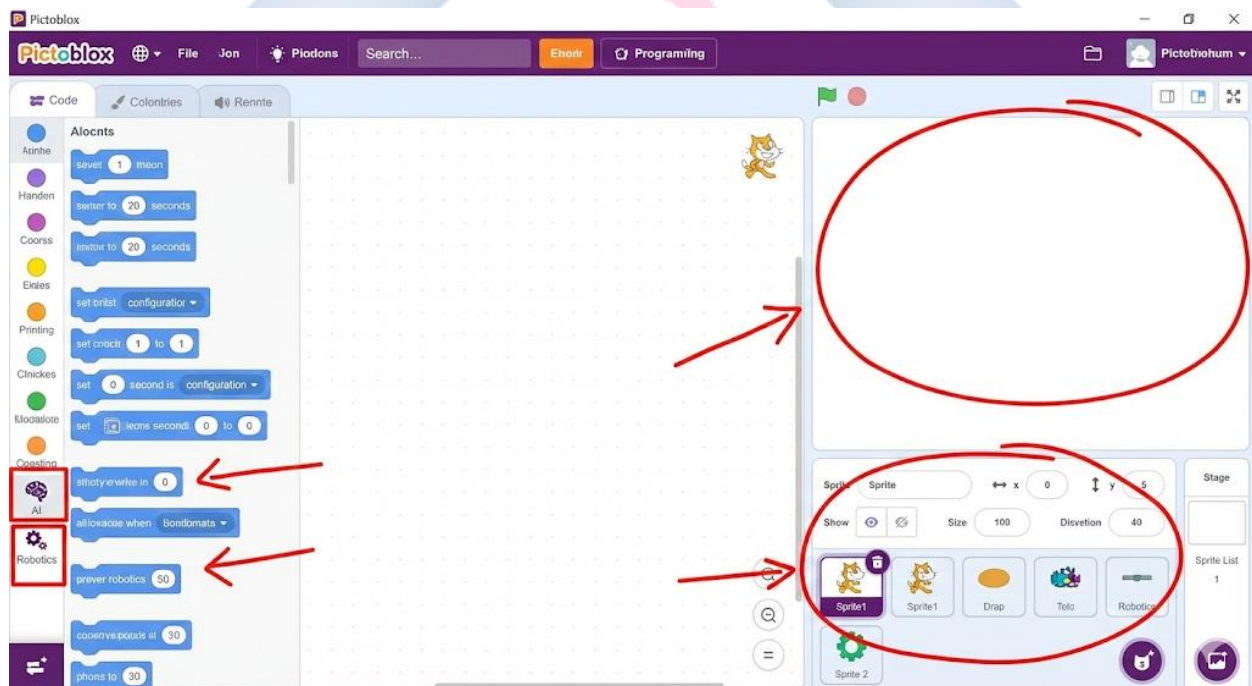
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Introduction to Pictoblox platform.
2. Set up the Pictoblox environment.
3. Create Sprites and Backdrops.
4. Implement basic game mechanics (movement and scoring).

CONTENT

1. Introduction to Pictoblox

Pictoblox is a powerful coding software based on **Scratch**. It has the same friendly, block-coding interface but includes special features for **Artificial Intelligence (AI)** and **Machine Learning**, and it can connect to and program real-life robots and hardware like the Arduino. It's excellent for making more advanced games and smart projects.



2. Setting Up the Environment

The Pictoblox workspace will look very familiar:

- **Stage:** Where your game happens.
- **Sprite List:** Where you choose and manage your characters.
- **Blocks Palette:** Color-coded blocks for coding (Motion, Looks, Sound, Events, **plus** special AI and Robotics blocks).
- **Coding Area:** Where you drag and snap blocks to create scripts.

3. Sprites and Backdrops

Just like in Scratch, you need characters and a world.

- **Choosing/Creating Sprites:** You can pick from the library, paint your own, or even use your **webcam to create a sprite from your own face or drawing!**
- **Choosing Backdrops:** Select from a library or paint a backdrop to set the scene for your game (e.g., a jungle, a city, space).



4. Basic Game Mechanics Let's build the core parts of a simple game, like "Catch the Falling Star":

- **Player Control:** Use when [right arrow v] key pressed and change x by (10) to move your main sprite (e.g., a basket) left and right.

- **Creating Objects:** Make a second sprite (e.g., a star) that uses a forever loop with glide to fall from the top of the stage.
- **Scoring System:**
 - a. Create a variable called Score.
 - b. Use a conditional: if <basket touching star?> then change [Score v] by (1).
 - c. Then, tell the star to go to a random position at the top to fall again.
- **Adding Sound:** Use start sound [Pop v] inside the if block to make a satisfying sound when you catch a star.

EVALUATION

1. Pictoblox is based on which simpler coding platform we already know?
2. Name one special feature Pictoblox has that standard Scratch does not emphasize.
3. What is a “Sprite” in Pictoblox?
4. Which block would you use to make the player-controlled sprite move to the right?
5. What do we create to keep track of points in a game?
6. Why is adding sound effects important in a game?

ASSIGNMENT

1. **Design:** On paper, draw the design for your own simple game. Sketch the **Hero** sprite and one **Enemy** or **Object** to collect.
2. **Written:** Write the Pictoblox code (in words) to make a sprite move **right** when the **right arrow key** is pressed.
3. **Creative:** List three different sound effects you would want in an action game (e.g., “boom” for an explosion, “ding” for scoring, “crash” for a collision).

WEEK 9: INTRODUCTION TO ARTIFICIAL INTELLIGENCE

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Artificial Intelligence (AI).
2. Differentiate between AI and Human Intelligence.
3. List three examples of AI in daily life.
4. Explain Machine Learning simply.

CONTENT

1. What is AI?

Artificial Intelligence (AI) is a part of computer science where we create **smart machines** that can **think, learn, and solve problems** in a way that seems human. “Artificial” means **man-made**. “Intelligence” means the **ability to learn and understand**.

- **Simple Definition:** AI is making computers **smart**.

2. AI vs. Human Intelligence

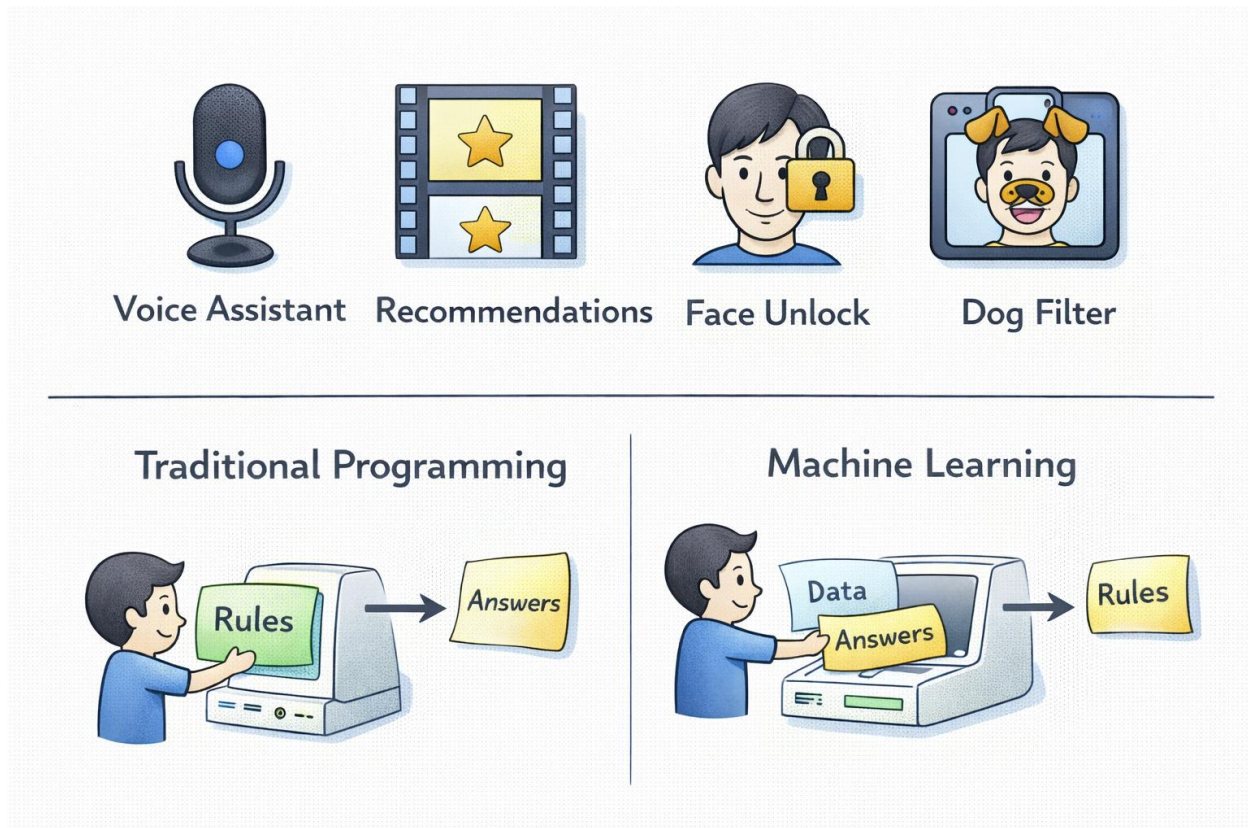
It's important to know that AI is different from human intelligence.

Human Intelligence	Artificial Intelligence (AI)
Can be creative (write poems, paint).	Follows patterns; not truly creative.
Has emotions (happy, sad).	Has no feelings or emotions.
Gets tired and needs rest.	Can work 24/7 without stopping.
Learns from a few experiences.	Needs thousands of examples to learn well.
Can understand context and jokes.	Takes everything literally.

3. AI in Daily Life You use AI more than you think! Here are common examples in Nigeria:

- **Voice Assistants: Siri (Apple), Google Assistant, Alexa.** You ask a question, they understand your voice and answer.

- **Recommendation Systems:** **YouTube** suggests videos you might like. **Netflix** suggests movies. **Jumia** suggests products. The AI learns from what you've watched or bought before.
- **Face Unlock:** Your parent's phone uses AI to recognize their face to unlock the screen.
- **Photo Apps:** When you use a filter on **Snapchat** or **Instagram** that puts dog ears on your face, AI finds your face first.



4. Machine Learning Basics Machine Learning (ML) is how most AI learns. It's the "learning" part of AI.

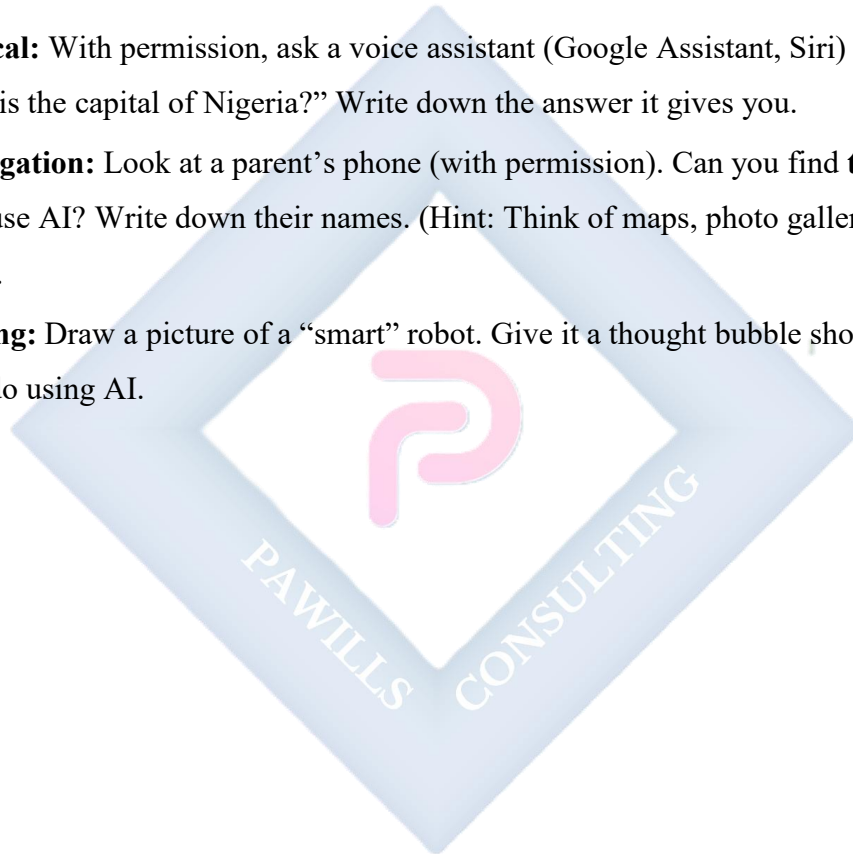
- **Traditional Programming:** Humans write all the rules. (If this, then that).
- **Machine Learning:** Humans give the computer **lots and lots of data (examples)** and the **correct answers**. The computer finds patterns in the data and **writes its own rules**. It's like showing a child 1,000 pictures of cats and dogs until they can recognize a new one on their own.

EVALUATION

1. What does “AI” stand for?
2. Name one thing a human can do that an AI cannot (from the table).
3. Give two examples of AI that you or your family might use on a phone.
4. How does YouTube know what kind of videos you like to watch?
5. What is Machine Learning?
6. True or False: An AI robot can feel happy when it wins a game.

ASSIGNMENT

1. **Practical:** With permission, ask a voice assistant (Google Assistant, Siri) a question like “What is the capital of Nigeria?” Write down the answer it gives you.
2. **Investigation:** Look at a parent’s phone (with permission). Can you find **two apps** that likely use AI? Write down their names. (Hint: Think of maps, photo galleries, social media).
3. **Drawing:** Draw a picture of a “smart” robot. Give it a thought bubble showing one task it could do using AI.



WEEK 10: AI APPLICATIONS AND ETHICS

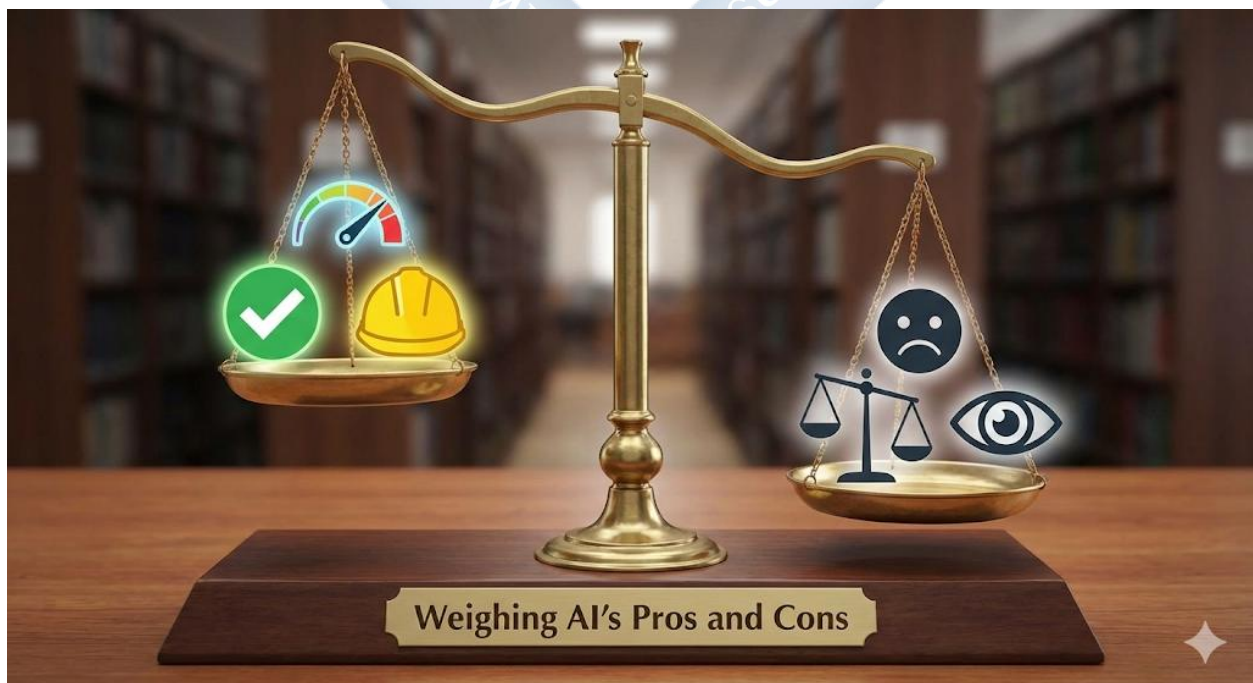
OBJECTIVES By the end of the lesson, pupils should be able to:

1. List three advanced applications of AI.
2. Mention two benefits of AI.
3. Identify concerns or ethical issues regarding AI.
4. Explain what it means to use AI responsibly.

CONTENT

1. Advanced AI Applications AI is doing amazing things in the world:

- **Self-Driving Cars:** Cars like **Tesla** use AI with cameras and sensors to see the road, obey traffic rules, and drive without a human driver.
- **Language Translation:** Apps like **Google Translate** use AI to instantly translate text or speech from English to Yoruba, Hausa, Igbo, or French.
- **Medical Diagnosis:** AI can help doctors by looking at X-ray or scan images and spotting signs of diseases (like tuberculosis or cancer) very quickly and accurately.
- **Smart Opponents:** The difficult enemies or opponents you play against in video games are often controlled by AI that learns how you play.



2. Benefits of AI

- **Efficiency & Speed:** AI can analyze huge amounts of data in seconds—work that would take humans years.
- **Accuracy:** In tasks like calculations or spotting flaws in products, AI makes very few mistakes.
- **Safety:** AI can do dangerous jobs (like exploring disaster sites) keeping humans out of harm's way.
- **Convenience:** AI powers helpful tools like traffic prediction in maps or voice-to-text messaging.

3. Concerns and Ethics (We Must Be Careful) With great power comes great responsibility.

There are worries about AI:

- **Job Displacement:** Some people worry that robots and AI will take over jobs from humans, like factory workers, drivers, or cashiers.
- **Bias and Unfairness:** If an AI is trained on bad or biased data, it will make unfair decisions. Example: An AI hiring tool might unfairly prefer one group of people over another if its training data was biased.
- **Privacy:** AI systems with facial recognition could be used to track people everywhere without their permission, which is an invasion of privacy.
- **Dependence:** Relying too much on AI might make some human skills weaker.

4. The Future: Using AI Responsibly The goal is to use AI as a **tool to help humans, not to replace or harm them**. Humans must always be **in control** and make the final important decisions.

- **We should build AI that is:** Fair, Transparent (we can understand how it works), and helpful to everyone.

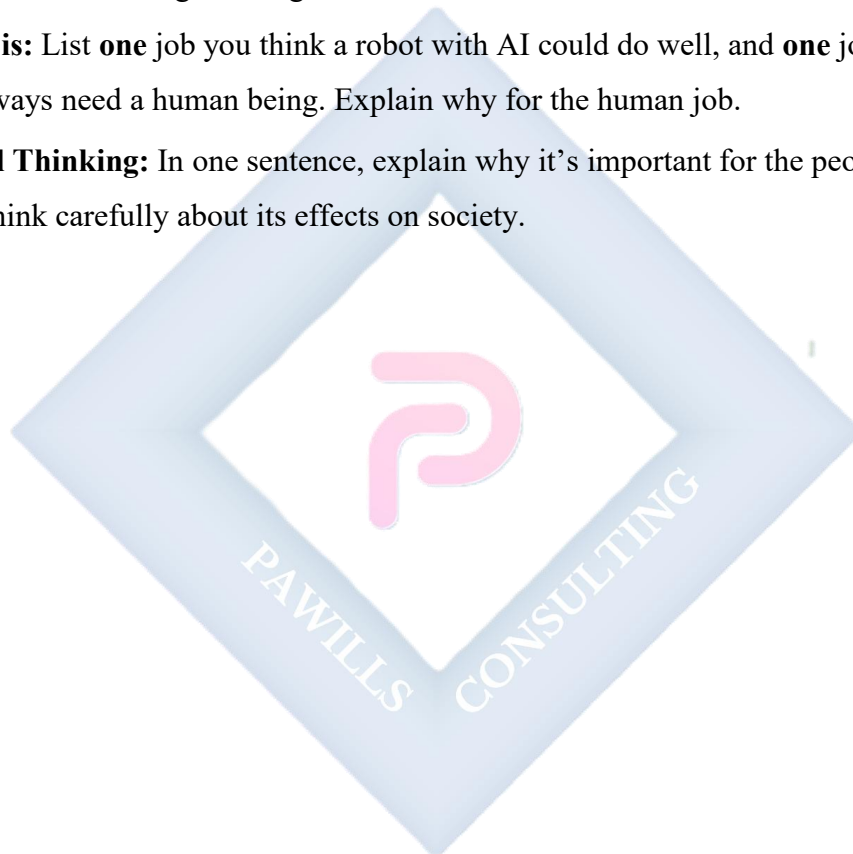
EVALUATION

1. Name one way AI is used in modern cars.
2. How does AI help people who speak different languages communicate?

3. What is one major **benefit** of using AI in factories?
4. Why are some people worried about AI and jobs?
5. What does “bias” in AI mean?
6. Who should ultimately be in charge of important decisions: a human or an AI system?

ASSIGNMENT

1. **Creative Writing:** Imagine the year 2040 in Lagos. Write 3-5 sentences describing what a ride in a self-driving taxi might be like.
2. **Analysis:** List **one** job you think a robot with AI could do well, and **one** job you think will always need a human being. Explain why for the human job.
3. **Ethical Thinking:** In one sentence, explain why it’s important for the people who create AI to think carefully about its effects on society.



WEEK 11: REVISION

CONTENT

1. Comprehensive Review This week, we will revisit the big ideas from the term:

- **Programming Concepts:** Sequence, Loops (repeat, forever), Conditionals (if-then, if-then-else).
- **Robotics:** Definition, parts (Sensors, Processor, Actuators), the “3 Ds,” and examples of robots.
- **Artificial Intelligence:** Definition, examples in daily life, Machine Learning, benefits, and ethical concerns.

2. Final Project: “Design a Helpful Robot” Pupils will work individually or in small groups on a creative planning project.

- **Task:** Design a robot that uses AI to solve a real problem in your community (e.g., in your school, home, or Lagos).
- **Presentation Plan:** Prepare to explain:
 - a. **Robot’s Name:** Give it a cool name.
 - b. **The Problem:** What will it solve? (e.g., picking up litter, helping elderly people carry things, tutoring in maths).
 - c. **How it Works:** What **sensors** will it need? (e.g., camera to see trash). What **actuators** will it use? (e.g., a gripper arm to pick up). How will **AI** help it? (e.g., learn to identify different types of trash).
 - d. **How it Helps:** Why is this robot a good idea for people?

EVALUATION

- Oral question-and-answer session covering key terms from the term.
- Assessment of the Final Project presentation based on clarity, creativity, and understanding of the concepts (robotics parts, AI role).

ASSIGNMENT

1. **Study:** Review your notes from Weeks 1-10. Make a list of 5 keywords from this term and their meanings.
2. **Prepare:** Finish your “Helpful Robot” design and presentation notes. Practice explaining it clearly.
3. **Completion:** Ensure any unfinished work in Pictoblox or Scratch from earlier weeks is completed and saved.

